

1-Adaptive Weak Pseudorandom Functions

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Abstract

A message authentication code (MAC) ensures authenticity and integrity in symmetric-key settings. The Carter–Wegman–Shoup (CWS) paradigm establishes that MACs for arbitrary-length messages can be built in a black-box way using a single call to a pseudorandom function (PRF) on a random input. More than a decade ago, Dodis, Kiltz, Pietrzak, and Wichs left open whether weak pseudorandom functions (wPRFs) would suffice in this construction.

This work establishes tight upper and lower bounds that precisely characterize the minimal computational assumptions needed for the security of the CWS paradigm. On the negative side, we prove that weak PRFs are insufficient to instantiate the CWS paradigm. On the positive side, we introduce a new primitive, the 1-adaptive weak pseudorandom function (1-awPRF), which guarantees pseudorandomness for polynomially many non-adaptive queries followed by one adaptive query. We show that 1-awPRFs are sufficient to secure CWS in a black-box manner.

Finally, we construct 1-adaptive weak pseudorandom functions in a black-box way from standard cryptographic assumptions, using a new randomized design paradigm that treats randomization as a fundamental structural element. Instantiating our generic construction under the Decisional Diffie Hellman and Learning with Errors assumptions yields concrete and efficient realizations. These lead to more efficient MAC schemes and illustrate how weak and abstract building blocks can be transformed into stronger and practically useful cryptographic constructions.

1 Introduction

A message authentication code (MAC) is the symmetric key analogue of a digital signature scheme, providing authenticity and integrity in two party settings. The seminal construction by Carter and Wegman [71], later refined by Shoup [68], combines an information theoretic hash function with a pseudorandom function (PRF):

$$\text{Mac}_{k_1, k_2}(m; r) = (H_{k_1}(m) \oplus F_{k_2}(r), r).$$

Here, the PRF is evaluated on a uniformly random value r , and its output serves as a one time pad that masks the hash value $H_{k_1}(m)$. This simple yet powerful design shows that a single call to a PRF suffices to construct a secure randomized MAC for messages of arbitrary length.

The Carter–Wegman–Shoup paradigm forms the foundation of many practical MACs and authenticated encryption schemes. Its structure appears in widely deployed designs such as GMAC

and POLY1305 (when instantiated with AES or CHACHA20), as well as in theoretical frameworks for one time and multi use MACs. The efficiency, modularity, and provable security of this paradigm make it one of the most enduring constructions in symmetric cryptography.

A key observation is that the PRF input in this construction is random rather than adversarially chosen. This raises the question of whether a weaker primitive could suffice. In particular, one may ask whether a *weak pseudorandom function (wPRF)*, which guarantees pseudorandomness only for random inputs, can replace the full PRF in the Carter–Wegman–Shoup paradigm. More than a decade ago, Dodis, Kiltz, Pietrzak, and Wichs posed this question as an open problem [32], and it has remained unresolved since.

This work investigates the minimal *complexity assumptions* needed to securely instantiate the CWS paradigm. On the negative side, we show that weak pseudorandomness is insufficient. On the positive side, we introduce a new primitive called the *1-adaptive weak pseudorandom function (1-awPRF)*, which bridges the gap between weak and full PRFs by allowing polynomially many non-adaptive queries on random inputs followed by a single adaptive query. We prove that 1-awPRFs are sufficient to establish the security of the CWS paradigm. Our main technical contribution is a construction of 1-awPRFs from strictly weaker primitives. Beyond the specific security proof, our construction departs from all prior designs by making essential use of randomness as a structural component rather than a source of unpredictability. This conceptual shift defines a new design paradigm that we believe is of independent interest.

As a byproduct, instantiating our 1-awPRF construction under the Decisional Diffie Hellman (DDH) and Learning with Errors (LWE) assumptions yields more efficient MACs. This demonstrates how theoretical insights into minimal complexity assumptions can translate into practical efficiency improvements by reducing the number of underlying cryptographic operations.

1.1 Our Results

Our main technical contribution is the construction of a 1-adaptive weak pseudorandom function that fundamentally differs from prior designs by embedding randomness as a structural element rather than treating it only as a source of unpredictability.

Theorem 1 (Informal). *Assuming the existence of pseudorandom generators, a 1-adaptive weak pseudorandom function exists.*

We then show that 1-adaptive weak pseudorandom functions suffice to securely instantiate the Carter–Wegman–Shoup paradigm.

Theorem 2 (Informal). *Instantiating the Carter–Wegman–Shoup paradigm with a 1-adaptive weak pseudorandom function and a XOR-universal hash function yields an unforgeable MAC. The latter can be upgraded to be unforgeable even in the presence of verification queries through minimal generic transformations.*

We further show that 1-adaptive weak pseudorandom functions are optimal in this hierarchy, whereas weak pseudorandom functions are not. Specifically, we construct explicit counterexamples demonstrating that certain combinations of universal hash families with weak pseudorandom functions yield insecure MACs. Some counterexamples are generic and do not rely on specific computational assumptions, while two of them are based on the DDH and LWE assumptions.

Theorem 3 (Informal). *There exist weak pseudorandom functions for which the Carter–Wegman–Shoup paradigm, even when combined with a strongly universal hash function, becomes forgeable.*

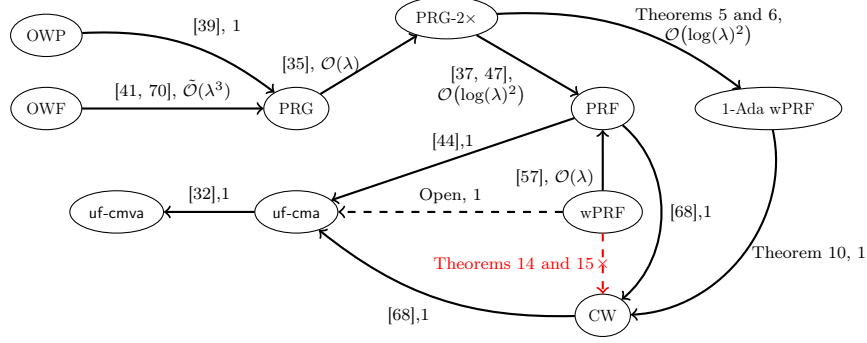


Figure 1: Relations among the primitives studied in this work. An edge from A to B labeled (S, N) indicates that B can be obtained from A via a black-box construction making N queries to A , with the corresponding proof given in source S . The red dashed edge with a cross marks the impossibility results established in this work. The label (Open, 1) denotes that obtaining a black-box single query construction remains an open problem. The PRG node represents the pseudorandom generator obtained from OWF and OWP, and PRG-2 $\times$ indicates a length-doubling PRG.

The existence of 1-adaptive weak pseudorandom functions establishes a clear gap between weak and full pseudorandom functions. Beyond this theoretical insight, it enables novel and highly efficient number-theoretic instantiations that reduce the number of cryptographic operations required for secure MAC constructions.

Theorem 4 (Informal). *Assuming the hardness of the Decisional Diffie Hellman or Learning with Errors problems, 1-adaptive weak pseudorandom functions can be constructed, yielding MACs that remain unforgeable even in the presence of verification queries.*

Finally, from a foundational perspective (cf. Figure 1), our results uncover an intriguing open problem. When instantiating our construction from length-doubling pseudorandom generators alone, the resulting MAC requires $\mathcal{O}(\log^2 \lambda)$ calls to the generator in a black-box manner. This bound matches that of Levin’s generic PRF construction $F_k(H(x))$, where $H: \{0, 1\}^\lambda \rightarrow \{0, 1\}^{\log^2 \lambda}$ is a universal hash function and F a small domain PRF [47]. Since a 1-adaptive weak pseudorandom function is strictly weaker than a full pseudorandom function, there should exist a construction achieving strictly fewer calls. Designing such a construction remains an open question for future work.

1.2 Related Work

The fundamental concept of pseudorandom functions (PRFs) was first introduced by Goldreich, Goldwasser, and Micali [38]. Weak pseudorandom functions (wPRF) are a similarly fundamental, related, and less strict variant of PRFs [65, 28]. There are multiple constructions of PRFs, including constructions from the decisional Diffie-Hellman problem [58], factoring [60], and lattices [7]. Additionally, there are constructions from the learning with errors problem (LWE) [67] and the learning parity with noise problem (LPN) [24]. Considering wPRFs, these constructions consider the even further relaxed definition of weak randomized PRFs of Applebaum et al. [5] and [50]. Due to LWE involving a much larger modulus than LPN, there exist several deterministic PRFs constructions [8, 6, 4, 25, 1, 45] based on the derandomized version of LWE, known as Learning With Rounding (LWR). These works leverage a rounding-based technique, which is not immediately applicable to

LPN. Nevertheless, a recent work by Ding et al. [31] showed a new independent derandomization technique for LPN.

Prior work considered other classes of keyed functions bridging between weak and full PRFs. *Non-adaptive* pseudorandom functions [56, 63, 64, 30] are a relaxation of PRFs where the adversarial queries are all chosen prior to seeing any PRF evaluation, but are not constrained to be uniformly random. Berman and Haitner [18] showed that non-adaptive PRFs can be cast to adaptive ones through a simple composition with a universal hash function. This class is known to contain full PRFs and is trivially a subset of weak PRFs.

The GGM construction [36] builds PRFs with input length $m = \text{poly}(\lambda)$ from m black-box queries to a length-doubling PRG. Levin’s domain extension technique [47] sharply reduced this to $\omega(\log \lambda)$ calls by preprocessing the input via a universal hash to outputs of bitlength $\omega(\log \lambda)$. Thanks to the works of Haitner, Reingold, and Vadhan [41] and Vadhan and Zheng [70], who showed that PRGs admit a black-box construction from one-way functions in $\tilde{O}(\lambda^3)$ calls, and the Goldreich-Levin Theorem [39], which implies a construction of PRGs from one-way permutations with exactly one call, one can derive a result on the query complexity with respect to the latter.

Despite several subsequent efforts towards improvements, this bound remains substantially the best to date. Maurer and Tessaro [52] present a technique for building PRFs from constant-query weak PRFs with a similar lower bound. Miles and Viola provided an initial lower bound [54] by proving the possibility of constructions consisting of projective single-calls, and strengthened it in [55] by ruling out constructions with non-adaptive calls and AC0 post-processing. More recently, Beimel, Malkin, and Mazon [9] proved the impossibility of constructing even weak PRFs through a generalized version of the GGM construction in $\mathcal{O}(\log \lambda)$ calls.

The numerous applications of PRFs in cryptography include encryption [12, 2], signatures [40, 14], and message authentication codes MAC [11, 10, 69], a symmetric key primitive providing authenticity and integrity. In particular, it is folklore that a PRF (with adequate image space size) directly yields a secure MAC scheme. Prominent examples for MAC are block cipher-based constructions like CBCMAC [16], ECBCMAC [15], XCBCMAC [22], TMAC [46], OMAC [43], as well as the parallelizable variants like PMAC [23] and LightMAC [49]. Besides that, there are different approaches to construction MACs including hash-based approaches like NMAC/HMAC [11, 10] and Poly1305-AES[19]. Especially the latter is interesting, since it follows the fundamental Carter-Wegman paradigm [71] for MACs. Essentially, these are one-time MACs which XOR the output of a strongly universal hash function with a secret nonce. Contemporary, this key is replaced by the output of a nonce-input, independently keyed PRF, and with a computation relaxation for the hash function [68, 27]. Black et al. [21] proposed a different combination strategy, where the universal hash’s output is fed to the PRF, possibly after appending a nonce [20]. However, utilizing a PRF is crucial yet stronger than necessary assumptions for these constructions (besides the one-time MACs). Dodis et al. [32] provide various constructions from different assumptions Alwen et al. [3] extend their results to construct key-indistinguishable MACs. Both papers provide constructions of MACs from key-homomorphic weak PRFs. Dodis et al. [32] as well as Daniel Masny [51] provide 3-round interactive authentication schemes from any weak PRF. However, obtaining a non-interactive construction from generic wPRFs remains an open problem.

1.3 Organization

Our work is organized as follows: Section 2 provides a self-contained technical outline; Section 3 introduces 1-adaptive weak PRFs, their black-box constructions, and separation from weak and full

PRFs; Section 4.2 proves that 1-adaptive weak PRFs enable secure Carter–Wegman–Shoup MACs, and analyzes the implications on query complexity; Section 5 contains proofs of the impossibility of black-box reductions from Carter–Wegman–Shoup MACs to weak PRFs; finally, Section 6 exhibits instantiations of the constructions from Section 3 based on the DDH and LWE assumptions.

For completeness, we provide relevant preliminary definitions and results in the Appendix: Section A contains definitions of weak and full pseudorandom functions, and universal hashes; Section B provides preliminaries on lattice-based cryptography and LWE; Section C lists and describes constructions of universal hashes and pseudorandom functions used throughout this work.

2 Technical Outline

2.1 Part I: Motivation and Impossibility

We study the minimal complexity assumption that allows us to prove the security of the Carter–Wegman–Shoup (CWS) paradigm [71, 68]

$$\text{MAC}_{k_1, k_2}(m; r) = (H_{k_1}(m) \oplus F_{k_2}(r), r),$$

where H_{k_1} is a XOR-universal hash and F_{k_2} is a PRF evaluated on a fresh random nonce r . This construction is provably secure, invokes the PRF on a bitstring whose length is independent of the message size, and requires only one PRF evaluation per tag.

Since the PRF is queried exclusively on uniformly random inputs, full pseudorandomness may be stronger than necessary. It is therefore natural to ask whether the PRF can be replaced by a *weak pseudorandom function* wF , which guarantees pseudorandomness only for random queries. Such a replacement could significantly reduce the overall cryptographic cost of the scheme. This raises the central question of our work:

What are the minimal assumptions to instantiate Carter–Wegman–Shoup securely?

Flawed Intuition: Weak PRFs on Uniform Inputs. A seemingly plausible argument for the security of a CWS MAC instantiated with a weak PRF goes as follows. For each message query m , the tag-generation algorithm samples a uniform nonce r and computes

$$t = H_{k_1}(m) \oplus \text{wF}_{k_2}(r).$$

Since wF is pseudorandom on uniform inputs, $\text{wF}_{k_2}(r)$ acts as a one-time pad masking the output of the XOR-universal hash $H_{k_1}(m)$. Moreover, the universality of H_{k_1} ensures that distinct messages collide only with negligible probability, suggesting that forgeries should be infeasible.

However, this reasoning fails in the unforgeability game. The adversary produces a forgery (m^*, t^*, r^*) such that m^* was never queried and Vrfy accepts. Crucially, the verification oracle does not restrict the distribution of r^* ; the adversary may freely choose it. Consequently, the weak PRF is no longer guaranteed to produce pseudorandom outputs, invalidating the above intuition. A simple counterexample illustrates this flaw. Let $H_{k_1}(x) := k_1 \cdot x$ be a universal hash, F_{k_2} be a weak PRF, and define

$$\text{wPRF}_{k_2}(r) := \begin{cases} F_{k_2}(r) & \text{if } r \neq 0, \\ 0 & \text{if } r = 0. \end{cases}$$

This function remains a valid weak PRF, as the event $r = 0$ occurs only with negligible probability under uniform sampling. Nevertheless, the resulting MAC is trivially insecure: the adversary can output $(m^*, t^*, r^*) = (0, 0, 0)$ as a valid forgery without any query. Hence, no black-box reduction can establish the security of CWS instantiated with arbitrary weak PRFs.

Insecurity under Algebraic Weak PRFs. One might suspect that the previous counterexample is somewhat artificial, arising only from the linearity of the XOR-universal hash used in the construction. A natural conjecture is that replacing it with a *strongly universal* (pairwise independent) hash function could eliminate such correlations. However, even this stronger choice does not restore security: the vulnerability stems from algebraic relations among the outputs of the weak PRF rather than from dependencies in the hash. The following counterexamples demonstrate that even with a strongly universal hash, the Carter–Wegman–Shoup paradigm remains insecure when instantiated with algebraically structured weak PRFs. We present two explicit examples, based respectively on the LWR and DDH assumptions, that illustrate this inherent fragility.

EXAMPLE 1: LWR-BASED WEAK PRF. Let the hash family over $(\mathbb{Z}_p, +, \cdot)$ be

$$H_{a,b}(x) = ax + b, \quad (a, b) \leftarrow \$ \mathbb{Z}_p^2,$$

and define the weak PRF derived from Learning With Rounding (LWR) as

$$\mathbf{wF}(k, r) = \lfloor \langle k, r \rangle \rfloor_p,$$

where $k \in \mathbb{Z}_q^n$, $r \in \mathbb{Z}_q^n$, $q > p$ prime, and $\lfloor a \rfloor_p := \lfloor a \cdot p/q \rfloor$. For suitable parameters, $\mathbf{wF}$ is a weak PRF under the LWE assumption [8, 6, 4, 25, 1, 45]. The tag for $m \in \mathbb{Z}_p$ and $r \leftarrow \$ \mathbb{Z}_q^n$ is

$$t = am + b + \lfloor \langle k, r \rangle \rfloor_p.$$

The LWR weak PRF satisfies an approximate additive homomorphism:

$$\mathbf{wF}(k, r_1) + \mathbf{wF}(k, r_2) = \mathbf{wF}(k, r_1 + r_2) - e, \quad e \in \{0, 1\}.$$

The adversary exploits this property and the affinity of $H_{a,b}$:

1. Query tags for $m_0 = 0$ and two distinct messages m_1, m_2 , obtaining $(t_0, r_0), (t_1, r_1), (t_2, r_2)$.
2. Compute $r^* = r_1 + r_2 - r_0$ and $t^* = t_1 + t_2 - t_0$.
3. Sample $\epsilon \leftarrow \$ \{-1, 0, 1\}$ and output $(m^*, t^*, r^*) = (m_1 + m_2, t^* + \epsilon, r^*)$.

By linearity,

$$t_1 + t_2 - t_0 = a(m_1 + m_2) + b + \mathbf{wF}(k, r_1 + r_2 - r_0) - e',$$

with $e' \in \{-1, 0, 1\}$. With probability $1/3$, $\epsilon = e'$, and the adversary succeeds.

EXAMPLE 2: DDH-BASED WEAK PRF. Similarly, let $(\mathbb{G}, +)$ be a cyclic group of prime order p , where DDH is hard. Define

$$H_{a,b}(x) = ax + b, \quad a \leftarrow \$ \mathbb{Z}_p, \quad b \leftarrow \$ \mathbb{G},$$

and the weak PRF

$$\mathbf{wF}(k, r) = k \cdot r$$

with $k \in \mathbb{Z}_p$ and $r \in \mathbb{G}$. This primitive is perfectly additive, enabling the same three-query combining attack without error guessing. Hence, the resulting MAC is completely forgeable.

TAKEAWAY. The above examples illustrate that the Carter–Wegman–Shoup paradigm, though optimal when instantiated with a full PRF, becomes fragile once the PRF is weakened or exhibits algebraic structure. Even under high-entropy randomness, algebraic relations between weak PRF outputs can be exploited to forge valid tags. The key insight is that all these attacks effectively introduce a *hidden adaptive query* to the weak PRF: when the adversary outputs a candidate forgery, the verification procedure itself evaluates the PRF on a new, adversarially chosen input. Hence, security requires pseudorandomness to hold not only for q random queries, but also for one additional adaptive query. This motivates an intermediate notion that captures precisely this minimal level of adaptivity, which we refer to as a *1-adaptive weak PRF*.

2.2 Part II: Towards 1-Adaptive Weak PRFs

Since weak pseudorandom functions are insufficient to prove the security of the Carter–Wegman–Shoup paradigm, we introduce a new primitive that lies strictly between weak and full PRFs: the *1-adaptive weak pseudorandom function*. Intuitively, a 1-adaptive weak PRF is a keyed function that remains computationally indistinguishable from a truly random function after the adversary has observed its outputs on polynomially many uniformly random inputs and is then allowed to make one adaptively chosen query. This definition aligns precisely with the security setting of the Carter–Wegman–Shoup construction, where the adversary has no control over the random nonces r used during tag generation and can only adaptively select a single input when attempting a forgery. The 1-adaptive weak PRF thus captures the minimal level of adaptivity required to model the verification phase of MAC security.

Warmup: A Lightweight Construction from Weaker Primitives. Our main technical contribution is a lightweight construction of a 1-adaptive weak PRF that avoids invoking a full PRF as a subroutine. We start from two weaker primitives: (i) a *small-domain PRF* S with polylogarithmic input lengths [36], and (ii) a λ -*bounded PRF* [33], which remains pseudorandom for up to λ adaptive evaluations. Our construction composes these primitives as

$$F(k, x||y) = B(S(k, y), x),$$

where $|x| = \lambda$ and $|y| = \log(\lambda)^2$. Conceptually, this design can be viewed as a single-layer tree, where each branch corresponds to one possible value of $S(k, y)$. Each branch has a budget of λ traversals, corresponding to evaluations of $B(S(k, y), \cdot)$. The key insight is that the randomly generated queries distribute almost uniformly among branches, ensuring that, with overwhelming probability, no branch exceeds its budget before the adaptive query. Formally, the suffix y selects an ephemeral key $z = S(k, y)$ for B , spreading the q random queries over $M = \lambda^{\log \lambda}$ independent buckets. For each fixed bucket j , the number of invocations follows a binomial distribution $\text{Bin}(q, 1/M)$, yielding an expected load of $q/M = q/\lambda^{\log \lambda}$, which for polynomial q is negligible relative to λ . Applying a Chernoff bound together with a Fréchet inequality, we show that, with overwhelming probability, no key is used more than $\lambda - 1$ times before the adaptive query, thereby preserving the λ -bounded pseudorandomness of B . This probabilistic load-balancing argument is the core technical insight enabling security under a single adaptive query while keeping the construction lightweight, relying only on one small-domain and one bounded PRF call.

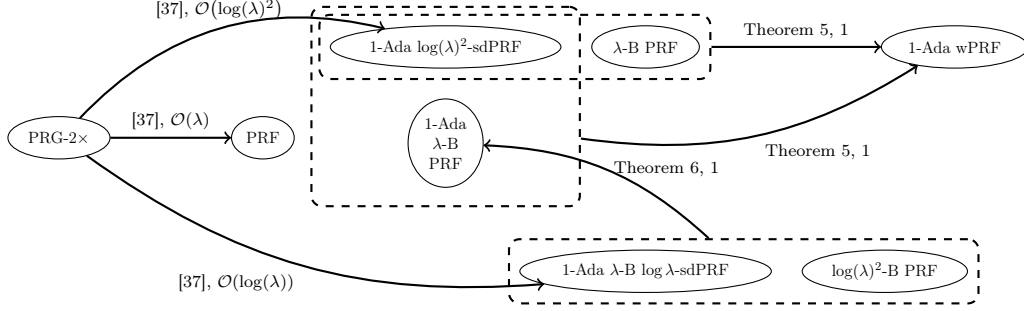


Figure 2: Overview of the constructions introduced in this work. The edge labels follow the semantics of Figure 1.

Note that the same design becomes insecure under fully adaptive queries: an adversary could fix a suffix $\bar{y}$ and repeatedly query on inputs sharing that suffix, thus exhausting the branch's budget in $\lambda + 1$ queries. Hence, our construction critically leverages the restricted query structure inherent in the 1-adaptive weak PRF model.

A More Lightweight B . While the above construction already minimizes dependence on strong primitives, it can be further optimized. In particular, the randomness of B 's input remains underutilized, suggesting that a fully bounded PRF provides stronger guarantees than necessary. It suffices that B tolerates up to $\lambda - 1$ random queries followed by a single adaptive one, yielding a natural relaxation of the 1-adaptive weak PRF notion with an explicit bound on the number of evaluations. We call such primitives *1-adaptive λ -bounded weak PRFs*. Because of their structural similarity to 1-adaptive weak PRFs, the previous analysis carries over. Concretely, using a small-domain PRF S with input size $\log \lambda$ and a $\log(\lambda)^2$ -bounded PRF C , we define

$$B(k, x \| y) = C(S(k, y), x),$$

and show that it satisfies 1-adaptive λ -bounded weak pseudorandomness under the same probabilistic arguments. Figure 2 summarizes this and other constructions proposed in this work.

A More Lightweight S . The same reasoning yields one final simplification. Analogous to B , we observe that in both the unoptimized and optimized constructions, S provides stronger guarantees than required. Specifically:

- In the first construction, S only needs to maintain pseudorandomness over q random inputs followed by one adaptively chosen query. Hence, it suffices for S to be a *small-domain 1-adaptive weak PRF*.
- The same holds for the second construction. Moreover, in that case S needs to support only $\lambda - 1$ random queries before the single adaptive one, that is, a *small-domain 1-adaptive λ -bounded weak PRF* suffices.

2.3 Part III: Applications

While our main technical contribution is primarily of theoretical interest, it naturally gives rise to two efficient instantiations, one under the DDH assumption and one based on the LWR problem.

Both constructions demonstrate that 1-adaptive weak PRFs can be instantiated from standard assumptions with significantly reduced key and computation complexity compared to their fully adaptive counterparts.

DDH-Based Instantiation. The DDH-based realization builds on the Naor–Reingold PRF [58]. We define S and S' with input sizes $\log(\lambda)^2$ and $\log(\lambda)$, respectively. Both are standard Naor–Reingold PRFs, where S' generates the $2d$ elements of $\mathbb{F}_Q$ forming the key of C . The resulting total key size is $O(\log(\lambda)^5 \lambda)$, compared to $O(\lambda^2)$ for a full Naor–Reingold PRF over λ -bit inputs. The evaluation cost is $O(\log(\lambda)^4 M(p) + \log(\lambda)^2 M(Q) + \log(\lambda)^2 A(Q))$, versus $O(\lambda M(p))$ for the standard PRF. By selecting reasonable multiplication and addition strategies from the literature, we effectively drop the time complexity from $\mathcal{O}(\lambda^2 \log \lambda)$ to $\mathcal{O}(\lambda \log(\lambda)^5)$.

Lattice-Based Instantiation. The LWR-based realization builds upon the LWR-based PRF of Boneh, Lewi, Montgomery, and Raghunathan [26]. We adapt this construction to the 1-adaptive weak setting by defining two auxiliary PRFs over small domains: S , which maps $\log(\lambda)^2$ input bits to elements of $\mathbb{Z}_{q_1}^m$, and S' , which maps $\log(\lambda)$ input bits to $\mathbb{Z}_{q_0}^m$, for appropriately chosen parameters q_0, q_1 , and m . The polylogarithmic input lengths of S and S' are consistent with the limitations on input size imposed by the security reduction in [26]. In contrast, employing the BLMR construction directly as a full-length PRF on inputs of size $\mathcal{O}(\lambda)$ would compromise security, as this would entail a reduction from an LWE instance known to be efficiently solvable. Our adapted instantiation preserves post-quantum security while enabling the use of existing lattice-based techniques that would otherwise be inapplicable.

3 p -Adaptive Weak Pseudorandom Functions

We first introduce a more general definition encompassing the idea of a keyed function whose output is pseudorandom when evaluated over arbitrarily many randomly generated inputs, and subsequently over a *bounded* number of adaptively chosen ones. By setting said bound to 1, one recovers the anticipated notion of 1-adaptive weak PRFs.

3.1 Definition

The notion of p -adaptive weak pseudorandom functions (p-awPRF) aims to bridge the gap between PRFs and wPRFs. In a PRF, all q input queries may be chosen adaptively, while in a wPRF, none are. A p-awPRF allows up to $p \leq q$ adaptive queries, thereby interpolating between these two extremes. We introduce the p -adaptive definition for the sake of generality and will use 1-adaptive weak PRFs throughout this work.

Definition 1 (p-adaptive wPRF). *We call $\text{pwF}: \mathcal{D}_\lambda \times \mathcal{D}_X \rightarrow \mathcal{D}_Y$ a (p) -awPRF if for all efficient distinguishers D , there exists a negligible function negl such that*

$$\left| \Pr \left[D(1^\lambda, \{(x_i, \text{pwF}(k, x_i))\}_{i=1}^q, \{(a_j, \text{pwF}(k, a_j))\}_{j=1}^p) = 1 \right] - \Pr \left[D(1^\lambda, \{(x_i, R(x_i))\}_{i=1}^q, \{(a_j, R(a_j))\}_{j=1}^p) = 1 \right] \right| \leq \text{negl}(\lambda),$$

where $(x_1, \dots, x_q) \leftarrow \$ \mathcal{D}_X$ are i.i.d. uniform inputs, and for polynomial many queries p with $j = 1, \dots, p$ the queries $a_j \in \mathcal{D}_X$ are chosen (possibly adaptively) by D after seeing previously returned pairs $\{(x_i, \mathbf{wF}(k, x_i))\}_{i=1}^q, \{(a_r, \mathbf{pwF}(k, a_r))\}_{r < j}$. The first probability is over a uniform key $k \in \mathcal{D}_\lambda$ and the randomness of D , while the second is over a uniform $R \in \text{Func}_y^x$ and the randomness of D .

3.2 Black-box Construction of 1-Adaptive wPRFs

Before diving into the technical aspects, we introduce two auxiliary classes of keyed functions at the core of our constructions. The first one is the class of *bounded pseudorandom functions*, i.e., functions that are indistinguishable from a random function for a bounded number of adaptive queries.

Definition 2 (Bounded PRF). *Fix $B \in \mathbb{N}$. A function $F: \mathcal{D}_\lambda \times \mathcal{D}_X \rightarrow \mathcal{D}_Y$ is B -bounded pseudorandom if for all efficient distinguishers D making at most B queries there exists a negligible function negl such that*

$$\left| \Pr \left[D^{F(k, \cdot)}(1^\lambda) = 1 \right] - \Pr \left[D^{f(\cdot)}(1^\lambda) = 1 \right] \right| \leq \text{negl}(\lambda),$$

where the first probability is over a uniform sampling of $k \in \mathcal{D}_\lambda$ and the randomness of D , while the second probability is over a uniform sampling of $f \in \text{Func}_y^x = \{f \mid f: \mathcal{D}_X \rightarrow \mathcal{D}_Y\}$ and the randomness of D .

The second class encompasses *adaptive bounded weak pseudorandom functions*, that is, weak PRFs that remain pseudorandom for a bounded number of queries on randomly sampled inputs, followed by a small number of adaptive queries.

Definition 3 (Adaptive Bounded Weak PRF). *Fix $B \in \mathbb{N}$. A function $\mathbf{pwF}: \mathcal{D}_\lambda \times \mathcal{D}_X \rightarrow \mathcal{D}_Y$ is a p -adaptive B -bounded PRF if for all efficient distinguishers D , there exists a negligible function negl such that*

$$\left| \Pr \left[D(1^\lambda, \{(x_i, F(k, x_i))\}_{i=1}^{B-1}, \{(a_j, \mathbf{pwF}(k, a_j))\}_{j=1}^p) = 1 \right] - \Pr \left[D(1^\lambda, \{(x_i, R(x_i))\}_{i=1}^{B-1}, \{(a_j, R(a_j))\}_{j=1}^p) = 1 \right] \right| \leq \text{negl}(\lambda),$$

where $(x_1, \dots, x_{B-1}) \leftarrow \$ \mathcal{D}_X$ are i.i.d. uniform inputs, and for polynomial many queries p with $j = 1, \dots, p$ the queries $a_j \in \mathcal{D}_X$ are chosen (possibly adaptively) by D after seeing previously returned pairs. The first probability is over a uniform key $k \in \mathcal{D}_\lambda$ and the randomness of D , while the second is over a uniform $R \in \text{Func}_y^x$ and the randomness of D .

Finally, we call a keyed function S a *small-domain PRF* if it is a pseudorandom function and has *polylogarithmic input length*. While being a subclass of PRFs, small-domain PRFs deserve particular attention due to being significantly more lightweight. For instance, instantiating them through a GGM tree requires polylogarithmically many evaluations of a PRG.

Black-box Construction We now exhibit a black-box construction based on the aforementioned tools. We anticipate that this initial construction shall be further optimized in a subsequent section. From a high-level viewpoint, we use two main building blocks, namely a λ -bounded PRF B and a small-domain PRF S with input of bitlength $\log(\lambda)^2$. At runtime, we break each input to the

candidate 1-adaptive weak PRF into a prefix x and a suffix y of length $\log(\lambda)^2$. Here, y is input for S to generate a key $z = S(k, y)$, which is used to evaluate $B(z, x)$. The latter is the final output returned by our construction.

Construction 1. Let $S: \mathcal{D}_\lambda \times \{0, 1\}^{\log(\lambda)^2} \rightarrow \hat{\mathcal{D}}_\lambda$ be a keyed function with input size $\log(\lambda)^2$ and key space $\mathcal{D}_\lambda$. Let $B: \hat{\mathcal{D}}_\lambda \times \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ be another keyed function with key space $\hat{\mathcal{D}}_\lambda$. Consider the keyed function $F: \mathcal{D}_\lambda \times \{0, 1\}^{\lambda + \log(\lambda)^2} \rightarrow \{0, 1\}^\lambda$ defined as

$$F(k, x||y) = B(S(k, y), x).$$

We briefly explain the rationale behind our design, as well as the intuition at the core of our proof of Theorem 5. By assumption, for each of the first q randomly generated inputs the y suffix is uniformly random over $\{0, 1\}^{\log(\lambda)^2}$, implying that each possible realization occurs with probability $\lambda^{-\log \lambda}$. If we turn to the corresponding ephemeral key $S(k, y)$, which is used to instantiate B , we know that the number of per-ephemeral-key usages follows a binomial distribution $\text{Bin}(q, \lambda^{-\log \lambda})$. This means, in turn, that each corresponding key $S(k, y)$ is used on average $q/\lambda^{\log \lambda}$ times. The mean value alone is rather meaningless, but we can leverage the Chernoff bound to prove that the probability of using the same key more than $\lambda - 1$ times is exponentially small in λ . While this is true for a fixed key, a union bound suffices to show that the probability that there exists one key that is used more than λ times is negligible in λ too. This preliminary statistical analysis enables a rigorous reduction to the security of S and B . At first, we replace S with a uniformly random function ρ_S . The resulting game is indistinguishable from that of Definition 1 by the pseudorandomness of S . We can also rule out collisions in ρ_S through a standard birthday bound. Next, we observe that B in the new game is evaluated over random keys, each of which is used less than λ times with high probability. This fully satisfies the constraints implicit in the definition of λ -bounded PRFs, allowing us to replace B with a uniformly random function, resulting in yet another a computationally indistinguishable game. More precisely, we replace every call of the type $B(z_i, \cdot)$ to a call to a fresh random function R_i , each time leveraging the pseudorandomness of B to argue indistinguishability. At this point, F 's outputs are uniformly random bitstrings by construction, and the proof is concluded.

Theorem 5. *If S is a small-domain PRF and B is a λ -bounded PRF, then Construction 1 is a 1-adaptive weak PRF.*

We first establish some preliminary notation and arguments before diving into the full proof. Let q denote the number of random queries to F , with $q \leq \lambda^d$ for a positive constant d . Let $x_i||y_i$ denote the i -th randomly generated query, and $z_i = S(k, y_i)$. Further, we introduce a vector C of size $\lambda^{\log \lambda}$ such that for each $j \in [\lambda^{\log \lambda}]$ the coordinate C_j contains the number of $i \in [q]$ such that $y_i = j$. Informally, C_j contains the number of times a call of the type $B(S(k, j), \cdot)$ was made. We now focus on C_1 for simplicity, but we note that all the following arguments also hold for any $j \in [\lambda^{\log \lambda}]$.

Proposition 1. $C_1 \sim \text{Bin}(q, 1/\lambda^{\log \lambda})$.

Proof. Since the inputs to F are uniformly random, the bitstrings $\{y_i\}_{i \in [q]}$ are random independent and identically distributed. Furthermore, for each fixed i the random variable $\delta(y_i, 1)^1$ is a Bernoulli with success probability $\Pr[y_i = 1] = 2^{-\log(\lambda)^2} = \lambda^{-\log \lambda}$. $\square$

¹Here δ denotes the Kronecker delta, which returns 1 if the inputs are equal, and 0 otherwise.

Proposition 2. *For each $d > 0$ such that $q \leq \lambda^d$ and sufficiently large λ we have*

$$\Pr[C_1 \geq \lambda] \leq e^{-\frac{\lambda-1}{2}}.$$

Proof. We start from the Chernoff bound for the binomial distribution, which states that if X is binomially distributed with mean μ and number of repetitions q . Then for any $\delta \geq 0$ we have

$$\Pr[X \geq (1 + \delta)\mu] \leq \exp(-\delta^2\mu/(\delta + 2)).$$

For C_1 , we get $\mu = \lambda^{d-\log \lambda}$. We now set $(1 + \delta)\mu = \lambda$, which leads to

$$\delta = \frac{\lambda}{\mu} - 1 = \lambda^{\log \lambda - d + 1} - 1.$$

For sufficiently large λ , this exceeds 0, and the bound is applicable. With $(1 + \delta)\mu = \lambda$, the argument of exp becomes

$$-\frac{\delta}{2 + \delta} \cdot (\delta\mu) = -\frac{\lambda^{\log \lambda - d + 1} - 1}{\lambda^{\log \lambda - d + 1} + 1} \cdot (\lambda - \lambda^{d - \log \lambda}).$$

For sufficiently large λ we get

$$\lambda - \lambda^{d - \log \lambda} \geq \lambda - 1,$$

$$\frac{\lambda^{\log \lambda - d + 1} - 1}{\lambda^{\log \lambda - d + 1} + 1} \geq 1/2.$$

Then the final bound

$$\Pr[C_1 \geq \lambda] \leq \exp(-(\lambda - 1)/2)$$

follows. □

Proposition 3. *For each $d > 0$ and sufficiently large λ we have*

$$\Pr[||C||_\infty < \lambda] \geq 1 - \frac{\lambda^{\log \lambda}}{e^{\frac{\lambda-1}{2}}}$$

Proof. Since for each $j \in [|C|]$

$$\Pr[C_j < \lambda] \geq 1 - e^{-\frac{\lambda-1}{2}}$$

by the Frechet inequality

$$\Pr \left[\bigwedge_{j=1}^{|C|} (C_j < \lambda) \right] \geq \sum_{j=1}^{|C|} \Pr[C_j < \lambda] - |C| + 1 \geq 1 - |C|e^{-\frac{\lambda-1}{2}}.$$

The statement in the proposition follows from the fact that $|C| = \lambda^{\log \lambda}$. □

In particular, we notice that $\frac{\lambda^{\log \lambda}}{e^{\frac{\lambda-1}{2}}}$ is negligible in λ . This means that, with overwhelming probability, during the 1-adaptive security game for each j , there are less than λ calls of type $B(j, \cdot)$. Thus, we never exceed the λ -bound of B and, regardless of $\mathcal{A}$'s choice for the final adaptive query $x||y$, the output remains pseudorandom.

Proposition 4. *If B is a keyed function with interfaces as defined in Construction 1, then $|\hat{\mathcal{D}}_\lambda| = \lambda^{\omega(1)}$.*

Proof (Sketch). Assume by contradiction that $|\hat{\mathcal{D}}_\lambda|$ is polynomial in λ . Then we exhibit an efficient distinguisher $D^\mathcal{O}$ with oracle access to either $\mathcal{O} = B(k, \cdot)$ or $\mathcal{O} = R(\cdot)$, where $k \leftarrow \$ \hat{\mathcal{D}}_\lambda$ and R is a uniformly random function:

1. D samples $\bar{k} \leftarrow \$ \hat{\mathcal{D}}_\lambda$.
2. D selects arbitrary pairwise distinct $x_1, \dots, x_{\lambda/2}$ and gets $y_i \leftarrow \mathcal{O}(x_i)$ for $i \in [\lambda/2]$.
3. D returns 0 iff $y_i = B(\bar{k}, x_i)$ for $i \in [\lambda/2]$.

Conditioned on the event $\hat{k} = k$, which has non-negligible probability, if $\mathcal{O} = B(k, \cdot)$ then D always returns 0. If $\mathcal{O} = R(\cdot)$, the probability of guessing the $\lambda/2$ evaluations of R is negligible. The details are left to the reader. $\square$

Proof of Theorem 5. Let us formalize the previous intuition through a reduction.

- **Game₀**: The real world implicit in the 1-adaptive security game according to Definition 1. That is, we play said game against an adversary $\mathcal{A}$, and always answer by honestly evaluating F .
- **Game₁**: This game is similar to **Game₀**, but the challenger replaces $S(k, \cdot)$ with a uniformly random function ρ_S . We argue that the games are indistinguishable by reducing to the pseudorandomness of S . Consider a generic PPT distinguisher D between **Game₀** and **Game₁**. Then we can efficiently use D to break the pseudorandomness of S as follows:

1. Let $\mathcal{R}^\mathcal{O}$ be a reduction with oracle access to either the small-domain PRF $S(k, \cdot)$ or the random function $\rho_S(\cdot)$.
2. $\mathcal{R}$ samples q uniformly random inputs $\{x_i||y_i\}_{i \in [q]}$, calls $\mathcal{O}(y_i) \rightarrow z_i$ and returns $B(z_i, x_i)$ to D .
3. At this point, D outputs an adaptive query $x||y$.
4. Then, $\mathcal{R}$ computes $z \leftarrow \mathcal{O}(y)$ and returns $B(z, x)$ to D .
5. Finally, D outputs its guess bit, and $\mathcal{R}$ outputs the same bit.

The reduction simulates the view of D perfectly and efficiently in all cases. Consequently, the PRF advantage directly transfers to $\mathcal{R}$ and we conclude

$$|\Pr[\text{Game}_0 = 1] - \Pr[\text{Game}_1 = 1]| \leq \text{Adv}_{\mathcal{R}}^{\text{PRF}, S}(\lambda)$$

- **Game₂**: This game is similar to **Game₁**, but the challenger aborts if at any point in time there exists an index j such that $C_j \geq \lambda$. This is statistically indistinguishable from **Game₁** by our bound on $\Pr[\|C\|_\infty < \lambda]$ given in Proposition 3. More precisely, for the difference between **Game₁** and **Game₂** we have

$$|\Pr[\text{Game}_1 = 1] - \Pr[\text{Game}_2 = 1]| \leq \lambda^{\log \lambda} \cdot e^{-(\lambda-1)/2}.$$

- **Game₃**: This game is similar to **Game₂**, but the challenger aborts if a collision for the output of the random function ρ_S occurs. More precisely, we abort if there exist two distinct $r, s \in \{y_i\}_{i \in [q]} \cup \{y\}$ such that $\rho_S(r) = \rho_S(s)$. Let $q_{Dist} = |\{y_i\}_{i \in [q]} \cup \{y\}|$ denote the number of distinct inputs to ρ_S . Through the standard birthday paradox, we get that the collision probability is at most $\frac{q_{Dist}^2}{|\mathcal{D}_\lambda|}$. Since $q_{Dist} \leq q + 1$

$$|\Pr[\text{Game}_2 = 1] - \Pr[\text{Game}_3 = 1]| \leq \frac{(q+1)^2}{|\mathcal{D}_\lambda|}.$$

- **Game₄**: We begin by stating some useful observations. Define the set $Q_Y = \{y_i\}_{i \in [q]} \cup \{y\}$, and

$$\rho_S(Q_Y) = \{\rho_S(y)\}_{y \in Q_Y}.$$

Intuitively, $\rho_S(Q_Y)$ contains all and only the distinct keys $z_i = \rho_S(y_i)$ (or $z = \rho_S(y)$) that are used to evaluate B . Due to the abort condition introduced in **Game₃**, conditioned on not aborting we get $|\rho_S(Q_Y)| = |Q_Y|$. Furthermore, because said condition rules out collisions, the number of queries of the type $B(\rho_S(j), \cdot)$ equals exactly C_j for each $j \in Q_Y$.

The difference to **Game₃** is that the challenger now replaces all calls of the type $B(z, \cdot)$ or $B(z_i, \cdot)$ with calls to uniformly random functions. More precisely, the challenger samples $|Q_Y|$ uniformly random functions $\{R_y\}_{y \in \rho_S(Q_Y \setminus \{y\})}, R$ and replaces the call to $B(z_i, \cdot)$ with a call to R_{y_i} (and the last call to $B(z, \cdot)$ with one to R). To formally prove that **Game₄** is indistinguishable from **Game₃**, we use a hybrid argument over the sequence of intermediate subgames **Game_{3,1}**, $\dots$, **Game_{3,q}**. In **Game_{3,1}** we keep everything as in **Game₃**, and only replace calls to $B(z_1, \cdot)$ with calls to a uniformly random function $R_1(\cdot)$ as described before². We bound the gap between **Game₃** and **Game_{3,1}** by reducing to the bounded pseudorandomness of B . Concretely, we exhibit a reduction $\mathcal{R}^\mathcal{O}$ with oracle access to $\mathcal{O}$, which is either a uniformly random function R_1 or $B(z_1, \cdot)$, with uniformly random z_1 . $\mathcal{R}$ runs a distinguisher D between **Game₃** and **Game_{3,1}** internally, and leverages D as follows:

1. $\mathcal{R}$ generates q uniformly random bitstrings $x_1 || y_1, \dots, x_q || y_q \leftarrow_{\$} \{0, 1\}^{\lambda + \log(\lambda)^2}$.
2. $\mathcal{R}$ computes $q - 1$ keys $z_2, \dots, z_q, z$ as follows: for $2 \leq i \leq q$, it computes $z_i \leftarrow \rho_S(y_i)$; it then computes $z \leftarrow \rho_S(y)$.
3. The reduction aborts if there is a collision in the $\{z_i\}_i \cup \{z\}$ values, as introduced in **Game₃**. Furthermore, it initializes a null counter $\text{ctr} = 0$ to track the number of queries of type $B(z_1, \cdot)$.

²We remark that if there exists a $i > 1$ such that $z_i = z_1$, the call $B(z_i, \cdot)$ should be replaced with a call to R_1 .

4. To issue the first random query to D , reduction $\mathcal{R}$ returns $(x_1||y_1, \mathcal{O}(x_1))$ and increments $\text{ctr} \leftarrow 1$.
5. For $1 < j \leq q$, reduction $\mathcal{R}$ issues the j -th random query to D as follows: if $y_i \neq y_1$, it returns $(x_j||y_j, B(z_j, x_j))$ using the z_i and x_i obtained at point 2; if $y_i = y_1$ and $\text{ctr} < \lambda - 1$, it increments ctr and returns $(x_i||y_i, \mathcal{O}(x_j))$; otherwise, $\mathcal{R}$ aborts.
6. When D makes the final adaptive query on input $x||y$, reduction $\mathcal{R}$ answers as follows: if $y \neq y_1$, it returns $(x||y, B(z, x))$; if $y = y_1$ and $\text{ctr} \leq \lambda - 1$, it returns $(x||y, \mathcal{O}(x))$; otherwise, $\mathcal{R}$ aborts.
7. Eventually, D outputs a decision bit and $\mathcal{R}$ outputs the same bit.

When $\mathcal{O}$ computes $B(z_1, \cdot)$, reduction $\mathcal{R}$ perfectly simulates **Game₃** for D . Similarly, when $\mathcal{O}$ computes $R_1(\cdot)$, reduction $\mathcal{R}$ perfectly simulates **Game_{3,1}**. Furthermore, notice that due to the abort condition on ctr , we never make more than λ queries to B . We now use the observations made from the beginning of **Game₄**: the abort conditions from **Game₂** and **Game₃** guarantee that, at this point, an abort due to large ctr never occurs. It follows that the gap between the games is upper-bounded by $\mathcal{R}$'s advantage in the bounded PRF game, that is

$$|\Pr[\text{Game}_3 = 1] - \Pr[\text{Game}_{3,1} = 1]| \leq \text{Adv}_{\mathcal{R}}^{\lambda-\text{PRF}, B}(\lambda).$$

More generally, for $i \geq 1$, **Game_{3,i+1}** is obtained from **Game_{3,i}** by replacing all calls to $B(z_{i+1}, \cdot)$ with calls to $R_i(\cdot)$. Through a reduction similar to the previous one, we get a bound

$$|\Pr[\text{Game}_{3,i} = 1] - \Pr[\text{Game}_{3,i+1} = 1]| \leq \text{Adv}_{\mathcal{R}}^{\lambda-\text{PRF}, B}(\lambda).$$

By a union bound over all the q games, we get

$$|\Pr[\text{Game}_3 = 1] - \Pr[\text{Game}_{3,q} = 1]| \leq q \cdot \text{Adv}_{\mathcal{R}}^{\lambda-\text{PRF}, B}(\lambda).$$

Finally, we transition from **Game_{3,q}** to **Game₄** by replacing the last query $B(z, x)$ for adaptively chosen x with a query to a uniformly random function R . Again, this is a reduction towards the bounded pseudorandomness of B , and we get

$$|\Pr[\text{Game}_{3,q} = 1] - \Pr[\text{Game}_4 = 1]| \leq \text{Adv}_{\mathcal{R}}^{\lambda-\text{PRF}, B}(\lambda).$$

Therefore, by the triangular inequality

$$|\Pr[\text{Game}_0 = 1] - \Pr[\text{Game}_4 = 1]| \leq \text{Adv}_{\mathcal{R}}^{\text{PRF}, S}(\lambda) + \lambda^{\log \lambda} \cdot e^{-(\lambda-1)/2} + \frac{(q+1)^2}{|\hat{\mathcal{D}}_\lambda|} + (q+1) \cdot \text{Adv}_{\mathcal{R}}^{\lambda-\text{PRF}, B}(\lambda).$$

In the final game, all q random queries and the final adaptive query are answered by evaluating uniformly random functions. On the other hand, in **Game₀** we perfectly simulate the real world of the 1-adaptive weak PRF game. Furthermore, in our setting, all additive terms on the right-hand side are negligible functions of λ . The advantages are negligible due to the pseudorandomness and bounded pseudorandomness of S and B , and $\lambda^{\log \lambda} \cdot e^{-(\lambda-1)/2}$ is inherently negligible. The λ -bounded pseudorandomness of B also implies that $|\hat{\mathcal{D}}_\lambda| = \lambda^{\omega(1)}$ (as shown in a previous proposition), making $(q+1)^2/|\hat{\mathcal{D}}_\lambda|$ negligible. This concludes the proof. $\square$

Similar to our first separating example, Construction 1 is not, in general, a full-fledged PRF. Suppose, for instance, that $\mathcal{A}$ is given oracle access to F and makes queries $x_1||0, \dots, x_{\lambda+1}||0$ for arbitrary but pairwise distinct inputs $x_1, \dots, x_{\lambda+1}$. This strategy causes the challenger to make $\lambda+1$ calls to $B(S(k, 0), \cdot)$ with no repeating inputs, thus overcoming the λ threshold and circumventing the guarantees of pseudorandomness provided by B .

3.3 Relaxing B to a 1-Adaptive Bounded Weak PRF

We note that the previous analysis of B relied only on the pseudorandomness of B against λ random queries and one final adaptive query. Given that the inputs x_i to B are uniformly random and independent (except for the last adaptive x), we can *lower the assumption* on B to be a *1-adaptive λ -bounded weak PRF*. The proof of Theorem 5 holds almost identically, with the reduction in the final hybrid game appealing to the 1-adaptive bounded weak pseudorandomness of B instead of full bounded pseudorandomness.

Construction 2. Let $S: \mathcal{D}_\lambda \times \{0, 1\}^{\log(\lambda)} \rightarrow \hat{\mathcal{D}}_\lambda$ be a keyed function with input size $\log(\lambda)$ and key space $\mathcal{D}_\lambda$. Let $C: \hat{\mathcal{D}}_\lambda \times \{0, 1\}^{\lambda - \log(\lambda)} \rightarrow \{0, 1\}^\lambda$ be another keyed function with key space $\hat{\mathcal{D}}_\lambda$. Consider the keyed function $B: \mathcal{D}_\lambda \times \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ defined as

$$B(k, x||y) = C(S(k, y), x).$$

Theorem 6. If S is a small-domain PRF, and C is a $\log(\lambda)^2$ -bounded PRF, then B is a 1-adaptive λ -bounded Weak PRF.

Proof. We adapt the notation of the previous section. We set $q = \lambda - 1$ to be the number of random queries to B , with $\{x_i||y_i\}_{i \in [q]}$ denoting the corresponding inputs and $x||y$ denoting the final adaptive query. Furthermore, let C denote a vector of length λ such that C_j denotes the number of i such that $y_i = j$. For each fixed j , $C_j \sim \text{Bin}(q, 1/\lambda)$ and its mean is $\mu_j = \frac{q}{\lambda} = 1 - 1/\lambda$. For sufficiently large (greater than or equal to 2) λ , $1 \geq \mu_j \geq 1/2$. By Chernoff's bound, for every $\delta > 0$

$$\Pr[C_j \geq 1 + \delta] \leq \Pr[C_j \geq \mu_j(1 + \delta)] \leq \exp\left(-\frac{\mu_j \delta^2}{2 + \delta}\right) \leq \exp\left(-\frac{\delta^2}{2(2 + \delta)}\right).$$

In particular, for $\delta = \ln(\lambda)^2 - 1$ and recalling that $\ln(x) < \log(x)$ for all real $x > 1$

$$\Pr[C_j \geq \log(\lambda)^2] \leq \Pr[C_j \geq \ln(\lambda)^2] \leq \exp\left(-\frac{(\ln(\lambda)^2 - 1)^2}{2(\ln(\lambda)^2 + 1)}\right).$$

We write the argument of exp as

$$-\frac{(\ln(\lambda)^2 + 1)^2 - 4\ln(\lambda)^2}{2(\ln(\lambda)^2 + 1)} = -\frac{\ln(\lambda)^2 + 1}{2} + 4\frac{\ln(\lambda)^2}{2(\ln(\lambda)^2 + 1)} \leq -\frac{\ln(\lambda)^2}{2} + 3/2.$$

Combining everything yields

$$\Pr[C_j \geq \log(\lambda)^2] \leq e^{3/2} \cdot \lambda^{-\ln \lambda/2}.$$

By a union bound over the λ coordinates of C we get

$$\Pr[\|C\|_\infty < \log(\lambda)^2] \geq 1 - \frac{e^{3/2}\lambda}{\lambda^{\ln \lambda/2}},$$

which is overwhelming in λ . Thus, when receiving the final adaptive query, with overwhelming probability there exists no $j \in [\lambda]$ such that $C(S(k, j), \cdot)$ received more than $\log(\lambda)^2 - 1$ queries. Consequently, regardless of the x the adversary provides, the pseudorandomness bound of C is not exceeded. Formally, we prove this by reduction similar to that of Theorem 5:

- **Game₀**: the real world implicit in the 1-adaptive λ -bounded security game according to Definition 3. That is, the challenger samples $\lambda - 1$ random inputs, honestly evaluates B over them, and returns the input-output pairs to $\mathcal{A}$. Then, it answers a single adaptive query.
- **Game₁**: This game is similar to **Game₀**, but the challenger replaces all the evaluations of $S(k, \cdot)$ with those of a uniformly random function ρ_S . The analysis of the difference between **Game₀** and **Game₁** is similar to the analysis of the difference between **Game₀** and **Game₁** in the proof of Theorem 5. The advantage of the corresponding reduction $\mathcal{R}$ bounds the gap between the two games, that is

$$|\Pr[\text{Game}_0 = 1] - \Pr[\text{Game}_1 = 1]| \leq \text{Adv}_{\mathcal{R}}^{\text{PRF}, S}(\lambda).$$

- **Game₂**: This game is similar to **Game₁**, but the challenger aborts if at any point there exists an index j such that $C_j \geq \log(\lambda)^2$. The difference between **Game₁** and **Game₂** emerges by our bound on the probability that $\|C\|_{\infty} < \log(\lambda)^2$, which is

$$|\Pr[\text{Game}_1 = 1] - \Pr[\text{Game}_2 = 1]| \leq e^{3/2} \frac{\lambda}{\lambda^{\ln \lambda/2}}.$$

- **Game₃**: Let $q_{Dist} = |\{y_i\}_{i \in [\lambda-1]} \cup \{y\}|$ denote the number of distinct inputs to ρ_S . By the standard birthday paradox, we get that the collision probability is at most $\frac{q_{Dist}^2}{|\mathcal{D}_{\lambda}|}$. Due to $q_{Dist} \leq \lambda$, we get

$$|\Pr[\text{Game}_2 = 1] - \Pr[\text{Game}_3 = 1]| \leq \frac{\lambda^2}{|\hat{\mathcal{D}}_{\lambda}|}.$$

- **Game₄**: This game is similar to **Game₃**, but we replace each call of the type $C(z, \cdot)$ and $C(z_i, \cdot)$ with a call to a uniformly random function. The details are identical to the transition from **Game₃** to **Game₄** in Theorem 5. Through a similar argument we conclude that

$$|\Pr[\text{Game}_3 = 1] - \Pr[\text{Game}_4 = 1]| \leq \lambda \cdot \text{Adv}_{\mathcal{R}}^{\lambda\text{-PRF}, C}(\lambda).$$

By the triangular inequality, we conclude

$$|\Pr[\text{Game}_0 = 1] - \Pr[\text{Game}_4 = 1]| \leq \text{Adv}_{\mathcal{R}}^{\text{PRF}, S}(\lambda) + e^{3/2} \frac{\lambda}{\lambda^{\ln \lambda/2}} + \frac{\lambda^2}{|\hat{\mathcal{D}}_{\lambda}|} + \lambda \cdot \text{Adv}_{\mathcal{R}}^{\lambda\text{-PRF}, C}(\lambda).$$

In **Game₄**, the challenger computes a uniformly random function and therefore, C 's output is uniformly random and independent. Furthermore, each term on the right-hand side is negligible in λ . This concludes the proof. $\square$

The previous theorem Theorem 6 enables an instantiation of B through simpler building blocks. S only parses $\log \lambda$ bits, and C only needs to support $\log(\lambda)^2$ adaptive queries, in contrast to the λ many queries required by B in Theorem 5.

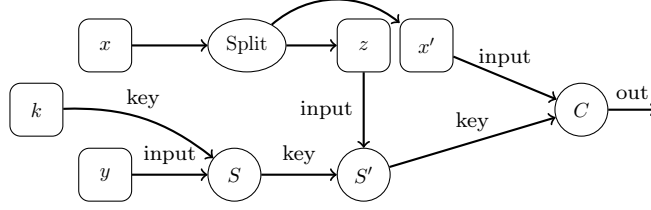


Figure 3: Visual representation of Construction 3.

Further Relaxations. We conclude the section with further observations and consequent optimizations. Similar to B in Theorem 5, C in Theorem 6 is required to stay pseudorandom when it is evaluated on $\log(\lambda)^2 - 1$ randomly sampled inputs, followed by a final adaptive query. Thus, it is sufficient to instantiate it as a 1-adaptive $\log(\lambda)^2$ -bounded weak PRF. Furthermore, the following proposition, which extends the above intuition and is used in a subsequent theorem, holds.

Proposition 5. *The conclusions of Theorem 5 and Theorem 6 hold even if S is relaxed to a small-domain 1-adaptive weak PRF or a 1-adaptive λ -bounded weak PRF, respectively.*

Proof (Sketch). Consider the proof of Theorem 5, and particularly **Game**₁, wherein we replace S with a uniformly random function ρ_S . We note that in **Game**₀, S is evaluated on q randomly generated inputs, followed by a final adaptively chosen one. Therefore, the transition to **Game**₁ remains legal if S is a 1-adaptive weak PRF. From that point on, we can recycle the proof of Theorem 5 as it is. Similarly, consider the proof of Theorem 6. We again note that in **Game**₁ we replace S with a uniformly random function ρ_S . At the same time, in **Game**₀ S is only evaluated over $\lambda - 1$ randomly generated inputs, followed by a single adaptively chosen one. Therefore, the step remains legal with a 1-adaptive λ -bounded weak PRF, and the rest of the proof remains unchanged. $\square$

3.4 Final Construction

By combining the initial Construction 1 with the speedup provided by Construction 2, we achieve the following generic construction, for which Figure 3 provides an overview.

Construction 3. *Let $S: \mathcal{D}_\lambda \times \{0, 1\}^{\log(\lambda)^2} \rightarrow \hat{\mathcal{D}}_\lambda$, $S': \hat{\mathcal{D}}_\lambda \times \{0, 1\}^{\log \lambda} \rightarrow \tilde{\mathcal{D}}_\lambda$, as well as $C: \tilde{\mathcal{D}}_\lambda \times \{0, 1\}^{\lambda - \log \lambda} \rightarrow \{0, 1\}^\lambda$ be three keyed functions. Then, we define another keyed function $F: \mathcal{D}_\lambda \times \{0, 1\}^{\lambda + \log(\lambda)^2} \rightarrow \{0, 1\}^\lambda$ according to the following procedure:*

- *Let $x||y$ be an input, with suffix y of length $\log(\lambda)^2$.*
- *Compute a key $k' \leftarrow S(k, y)$.*
- *Write $x = x'||z$, where z is a suffix of length $\log(\lambda)$.*
- *Output $C(S'(k', z), x')$.*

Theorem 7. *Consider Construction 3 instantiated as follows: S is a small-domain 1-adaptive weak PRF; S' is a small-domain 1-adaptive λ -bounded weak PRF; C is a $\log(\lambda)^2$ -bounded PRF. Then F is a 1-adaptive weak PRF.*

Proof. By Theorem 5, we know how to combine small-domain PRF S and a λ -bounded PRF B to build a 1-adaptive weak PRF. By Theorem 6, we further know how to construct the latter from a small-domain PRF S' and a $\log(\lambda)^2$ -bounded PRF C . It is straightforward to verify that Construction 3 combines these building blocks as prescribed by both theorems. Finally, by Proposition 5, we can relax S and S' to be a 1-adaptive weak PRF and a 1-adaptive λ -bounded weak PRF, respectively. $\square$

We now discuss some of the possible choices to instantiate Construction 3. We intentionally keep the discussion informal here, and defer concrete instantiations of the building blocks based on the DDH and the LWR assumption to Section 6.

Instantiating S . We can construct S in a black-box way from any length-doubling pseudorandom generator PRG through the GGM construction [36]. Due to the short input length of S , it is sufficient to iterate the underlying PRG a total of $\log(\lambda)^2$ times. Alternatively, we can instantiate S from ad-hoc computational assumptions. Notable examples include, but are not limited to, the Naor-Reingold DDH-based PRF [58], and any of the LWR-based PRFs available in the literature [8, 6, 26].

Instantiating B and C . We can instantiate B with a λ -wise independent hash function, which offers an information-theoretic approach but demands an impractically large key (e.g., $\mathcal{O}(\lambda^2)$ for a degree $\lambda - 1$ polynomial). A more practical method, enabled by Construction 2, instead reduces B to a small-domain PRF S' and a $\log(\lambda)^2$ -bounded PRF C , allowing the latter to be efficiently instantiated as a $\log(\lambda)^2$ -wise independent hash. Alternative constructions also exist, such as the black-box construction by Döttling and Schröder [33] using a small-domain PRF with universal hashes. Besides black-box constructions, standard ad-hoc assumptions like Learning With Rounding (LWR) can ensure bounded pseudorandomness. While LWR samples are known to provide polynomial unbounded pseudorandomness when q is exponential in the security parameter, switching to polynomial q restricts the number of consecutive samples that are guaranteed to be pseudorandom [4, 25, 1, 45].

A Note on the Security Parameter. In the proof of proposition 2 we leverage the fact that for sufficiently large λ we get

$$\lambda - \lambda^{d-\log \lambda} \geq \lambda - 1,$$

$$\frac{\lambda^{\log \lambda - d + 1} - 1}{\lambda^{\log \lambda - d + 1} + 1} \geq 1/2.$$

In a concrete instantiation, we must select λ large enough to ensure both. To satisfy the first condition, we need that $d \leq \log \lambda$, which in turn implies $\lambda \geq 2^d$ to tolerate λ^d many queries. For this example, let $q = 2^{100}$ be an arbitrary but cryptographically sufficiently large q . This results in the inequality $100 \leq (\log \lambda)^2$, which implies $\lambda \geq 2^{10} = 1024$. When equality holds, the second condition becomes

$$\frac{\lambda^{\log \lambda - d + 1} - 1}{\lambda^{\log \lambda - d + 1} + 1} = \frac{\lambda - 1}{\lambda + 1} = 1023/1025 \geq 1/2.$$

To satisfy the first condition, we require $d \leq \log \lambda$. If we assume the query complexity q (from Proposition 1) is bounded by $q \approx \lambda^d$, this implies the constraint $\log q \leq (\log \lambda)^2$. For a concrete

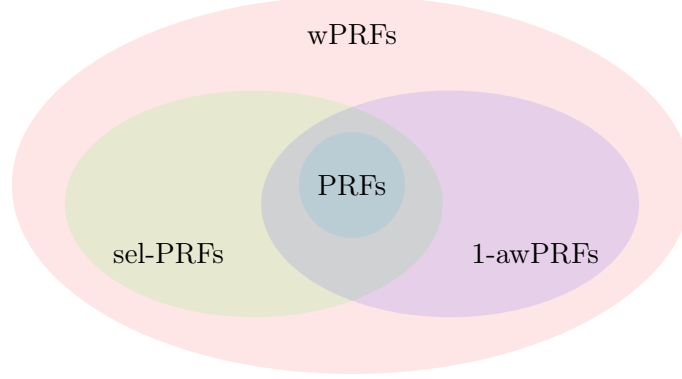


Figure 4: Known containment relations between classes of pseudorandom functions.

example like $q = 2^{100}$, this is satisfied by $\lambda = 1024$, which also satisfies the second condition and confirms the practical applicability of the bounds used in the proof of Proposition 2.

3.5 Separating Examples

In the following, we prove containment relations between awPRFs and related classes of pseudorandom functions. The results are summarized in Figure 4.

It is trivial to verify that every 1-adaptive weak PRF is also a weak PRF, and that full PRFs are also 1-adaptively weak. The previous constructions also show that the class of 1-adaptive weak PRFs is truly an intermediate one, in the sense that it does not coincide with that of weak or full PRFs. This section provides a much simpler separating example proving that PRFs are a strict superset of 1-adaptive weak PRFs. That is, we show that there exists a 1-adaptive weak PRF that is not a PRF. The example's simplicity comes at a high computational cost, due to running an internal PRF. Therefore, the associated construction should only be viewed by means of conceptually simplifying the claimed separation. Furthermore, this separating example turns out to be usable to prove yet another separation result, this time between 1-adaptive weak PRFs and non-adaptive PRFs.

Construction 4. Consider a keyed function $F: \mathcal{D}_\lambda \times \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$. We define a new keyed function F as

$$F(k_0, k_1, x) = \begin{cases} F(k_0, x) & x \equiv 0 \pmod{3} \\ F(k_1, x - 1) & x \equiv 1 \pmod{3} \\ F(k_0, x - 2) \oplus F(k_1, x - 2) & x \equiv 2 \pmod{3} \end{cases},$$

where $x \pmod{3}$ is computed by parsing x as the encoding of an integer between 0 and $2^n - 1$.

Theorem 8. If F is a PRF and k_0 and k_1 are uniformly random and sampled independently, then Construction 4 is a 1-adaptive weak pseudorandom function, but not a full-fledged pseudorandom function.

Proof. The second part of the claim is easily shown by exhibiting the following PPT distinguisher $D^\mathcal{O}$ with oracle access to $\mathcal{O}$, which can be either $F(k_0, k_1, \cdot)$ or a uniformly random function f :

1. Obtain three samples $y_i \leftarrow \mathcal{O}(i)$ for $i = 0, 1, 2$.

2. Return 1 if $y_1 \oplus y_2 = y_3$, and 0 otherwise.

When $\mathcal{O} = F(k_0, k_1, \cdot)$, you return 1 with probability 1 by construction. When $\mathcal{O} = f(\cdot)$, y_1, y_2 and y_3 are uniformly random independent values, and $\Pr[y_1 \oplus y_2 = y_3] = 2^{-\lambda}$.

We now turn to the first claim. Consider the following sequence of games:

- **Game₀**: The game implicit in the definition of 1-adaptive weak PRFs, where D must distinguish between a uniformly random function f , and a $F(k_0, k_1, \cdot)$ with $k_0, k_1 \leftarrow \$\mathcal{D}_\lambda$ and independent.
- **Game₁**: Let q denote the number of non-adaptive queries to the oracle. We modify the previous game by checking if there exists a pair of distinct indices $i, j \in [q]$ such that $\lfloor x_i/3 \rfloor = \lfloor x_j/3 \rfloor$, and aborting if that is the case. We call the former conditions C . For exactly two random samples, the probability for an abort is at most $2/2^\lambda$. Through a union bound over the possible $\binom{q}{2} \leq q^2$ pairs, we get an upper bound of $\frac{2q^2}{2^\lambda}$. As a result, we have

$$|\Pr[\text{Game}_0 = 1] - \Pr[\text{Game}_1 = 1]| \leq \frac{2q^2}{2^\lambda}.$$

- **Game₂**: We modify the previous game by replacing $F(k_0, \cdot)$ with a uniformly random function R_0 . That is, whenever the challenger receives a call that requires an evaluation of $F(k_0, \cdot)$, it instead evaluates a uniformly random function $R_0(\cdot)$. To show the indistinguishability between **Game₁** and **Game₂** we provide a reduction $\mathcal{R}$ to the pseudorandomness of F that leverages a distinguisher D between the games:

1. $\mathcal{R}^\mathcal{O}$ has access to an oracle $\mathcal{O}$, which equals either $F(k_0, \cdot)$ or $R_0(\cdot)$.
2. $\mathcal{R}$ samples a uniformly random k_1 .
3. Upon receiving a query x_i chosen at random, $\mathcal{R}$ replies as follows: if the condition C is violated, it aborts; if $x_i \equiv 0 \pmod 3$, $\mathcal{R}$ returns $\mathcal{O}(x_i)$; if $x_i \equiv 1 \pmod 3$, $\mathcal{R}$ returns $F(k_1, x_i - 1)$; otherwise, $\mathcal{R}$ returns $\mathcal{O}(x_i - 2) \oplus F(k_1, x_i - 2)$.
4. Upon receiving the final adaptive query x from D , $\mathcal{R}$ replies as follows: if $x \equiv 0 \pmod 3$, $\mathcal{R}$ returns $\mathcal{O}(x)$; if $x \equiv 1 \pmod 3$, $\mathcal{R}$ returns $F(k_1, x - 1)$; otherwise, $\mathcal{R}$ returns $\mathcal{O}(x - 2) \oplus F(k_1, x - 2)$.
5. At some point, D outputs a guess bit b' . $\mathcal{R}$ outputs b' .

If $\mathcal{O}$ equals $F(k_0, \cdot)$, reduction $\mathcal{R}$ simulates **Game₁** perfectly for D . Similarly, if $\mathcal{O}$ equals R_1 then $\mathcal{R}$ perfectly simulates **Game₂**. Furthermore, the simulation is efficient. As a result, we have

$$|\Pr[\text{Game}_1 = 1] - \Pr[\text{Game}_2 = 1]| \leq \text{Adv}_{\mathcal{R}}^{\text{PRF}, F}(\lambda).$$

- **Game₃**: We modify the previous game by replacing $F(k_1, \cdot)$ with a uniformly random function R_0 . That is, whenever the challenger receives a call that requires an evaluation of $F(k_1, \cdot)$, it instead evaluates a uniformly random function $R_1(\cdot)$. The modification is identical to the one defining **Game₂**, and by a similar reduction $\mathcal{R}'$ to the pseudorandomness of F we conclude

$$|\Pr[\text{Game}_2 = 1] - \Pr[\text{Game}_3 = 1]| \leq \text{Adv}_{\mathcal{R}'}^{\text{PRF}, F}(\lambda).$$

We now analyze the last game. Notice that the abort condition C implies another condition, namely that for any pairwise distinct indices i, j, ℓ it never holds that $\lfloor x_i/3 \rfloor = \lfloor x_j/3 \rfloor = \lfloor x_\ell/3 \rfloor$. We call the latter condition C' . With this in mind, we define two sets A and B as

$$A = \{\lfloor x_i/3 \rfloor\}_{i \in [q]} \quad \text{and} \quad B = \{x_i \bmod 3\}_{i \in [q]}.$$

Furthermore, we relabel $a_i = \lfloor x_i/3 \rfloor$, $b_i = x_i \bmod 3$, and $R_2(x) = R_0(x) \oplus R_1(x)$ for all k and x . In the last game the output y_i corresponding to x_i equals

$$y_i = F(k_0, k_1, x_i) = R_{b_i}(a_i) \quad \forall i \in [q].$$

Let x denote the final adaptive query, and let $a = \lfloor x/3 \rfloor$ and $b = x \bmod 3$. We enumerate all the possible cases:

- $x = x_i$ for some $i \in [q]$: then, by C' holding, the $\{y_i\}_{i \in [q]}$ values are uniformly random and independent, and the final adaptive query merely returns one of them.
- $a \in A$, but if $a = a_i$ for some i then $b \neq b_i$: suppose i is the unique index such that $a_i = a$. The claim of uniqueness is justified by the fact that, due to condition C , for any $j \neq i$ it must hold $a_i \neq a_j$. By construction, the corresponding output y equals $R_b(a)$. By assumption, since $b \neq b_i$, distinguisher D got either $R_{b+1 \bmod 3}(a)$ or $R_{b+2 \bmod 3}(a)$. Without loss of generality, suppose it got $R_{b+1 \bmod 3}(a)$. The values $y = R_b(a)$ and $y_i = R_{b+1 \bmod 3}(a)$, together with the rest of the transcript's outputs, are uniformly random and independent.
- $a \notin A$: then $y = R_b(a)$ is the output of R_b at a fresh input a . Therefore, together with the rest of the y_i variables, it is uniformly random and independent.

By triangular inequality $|\Pr[\text{Game}_0 = 1] - \Pr[\text{Game}_3 = 1]| \leq \text{Adv}_{\mathcal{R}}^{\text{PRF}, F}(\lambda) + \text{Adv}_{\mathcal{R}'}^{\text{PRF}, F}(\lambda) + \frac{2q^2}{2^\lambda}$. This concludes the proof. $\square$

Remark 1. *The counterargument in the previous proof is defined over $(\{0, 1\}^\lambda, \oplus)$ for simplicity, but would work similarly for any finite additive group $(\mathbb{G}, +)$.*

Theorem 8 implicitly shows yet another separation result, i.e., between awPRFs and non-adaptive PRFs. We note that for a non-adaptive PRF, a distinguisher has to choose and fix all queries in advance (c.f. Section A)

Proposition 6. *There exists a function that is 1-adaptive weak pseudorandom, but not selectively pseudorandom.*

Proof. Consider Construction 4 instantiated as in Theorem 8. We already showed that the result is a 1-awPRF. Furthermore, we observe that the proof of Theorem 8 exhibits an attack that is selective. $\square$

The converse is true as well: there exists a PRF that is selectively secure, but not a 1-awPRF. The proof of this result is considerably more involved and requires a new construction.

Construction 5. *Let $P: \mathcal{D}_\lambda \times \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ be a 1-to-1 keyed function. Define a new function $F: \mathcal{D}_\lambda \times \{0, 1\}^{\lambda+1} \rightarrow \{0, 1\}^\lambda$ as follows*

$$F(k, x||b) = \begin{cases} P(k, x) & b = 0 \\ P^{-1}(k, x) & b = 1 \end{cases}.$$

We anticipate that Construction 5, instantiated with a *strong* pseudorandom permutation (c.f. Section A), yields a pseudorandom function that is selectively secure, not secure with one adaptive query. From an intuitive viewpoint, a 1-adaptive adversary can issue a query to F in such a way that the mapping P on a previously seen query is inverted. On the other hand, a selective adversary cannot do it without previously seeing F 's outputs, as it lacks information on the bitstring where P should be inverted. Finally, we have to resort to strong pseudorandom permutations as $\mathcal{A}$ is implicitly given oracle access to P and its inverse simultaneously. We remark that the latter exist based on the assumptions that pseudorandom functions exist, and can be efficiently constructed from the latter through a four-round Feistel network, as proven by Luby and Rackoff [48, 59].

Theorem 9. *If P is a strong pseudorandom permutation, then Construction 5 is a non-adaptive PRF, but NOT a 1-awPRF.*

Proof(Sketch). We start by proving the second claim. Let $r||b$ denote a randomly generated input to F (from Construction 5). An efficient adversary receives $r||b, F(k, r||b)$, queries F adaptively on input $F(k, r||b)||-b$, and checks if the output equals r . This works because on input $F(k, r||b)||-b$ the oracle returns

$$P^{(-1)^{-b}} \left(k, P^{(-1)^b}(k, r) \right) = r.$$

This event has negligible probability when the oracle evaluates a uniformly random function, instead. Consequently, F is not a 1-awPRF.

We now turn to the first claim. Let $x_1||b_1, \dots, x_q||b_q$ denote the adversarially generated inputs. Without loss of generality, assume that there exists an $\ell \in [q]$ such that $b_1 = \dots = b_\ell = 0$, and $b_{\ell+1} = \dots = b_q = 1$. By construction, the adversary receives evaluations

$$P(k, x_1), \dots, P(k, x_\ell), P^{-1}(k, x_{\ell+1}), \dots, P^{-1}(k, x_q).$$

By the pseudorandomness of P , this is computationally indistinguishable from

$$\pi(x_1), \dots, \pi(x_\ell), \pi^{-1}(x_{\ell+1}), \dots, \pi^{-1}(x_q),$$

where π denotes a uniformly random permutation. Now, consider a tuple $(y_1, \dots, y_q) \in (\{0, 1\}^\lambda)^q$ denoting a possible outcome of the previous random variable. We can obtain said tuple iff

$$\pi(x_1), \dots, \pi(x_\ell), \pi(y_{\ell+1}), \dots, \pi(y_q) = y_1, \dots, y_\ell, x_{\ell+1}, \dots, x_q.$$

To analyze the collision probabilities that distinguish these outputs from purely random strings, we consider the following sequence of games:

- **Game₀**: This game samples a uniformly random permutation Π , and returns $\{\Pi(x_i)\}_{i \in [q]}$.
- **Game₁**: This game samples a uniformly random permutation Π . Subsequently, it checks if $\{\Pi(x_1), \dots, \Pi(x_\ell)\} \cap \{x_{\ell+1}, \dots, x_q\} = \emptyset$. If the condition holds, it returns $\Pi(\vec{x})$; otherwise, it aborts. Note that **Game₀** and **Game₁** proceed identically unless the abort condition is triggered. The probability of aborting equals the probability of guessing either of $\Pi(x_1), \dots, \Pi(x_\ell)$, which is upper bounded by $\frac{\ell}{2^\lambda}$. Therefore:

$$|\Pr[\text{Game}_0 = 1] - \Pr[\text{Game}_1 = 1]| \leq \frac{\ell}{2^\lambda}.$$

- **Game₂**: This game samples a uniformly random permutation Π , and check if $\{x_1, \dots, x_\ell\} \cap \{\Pi^{-1}(x_{\ell+1}), \dots, \Pi^{-1}(x_q)\} = \emptyset$. If so, it returns $\Pi(x_1), \dots, \Pi(x_\ell), \Pi^{-1}(x_{\ell+1}), \dots, \Pi^{-1}(x_q)$; otherwise, it aborts. We claim that the games are perfectly indistinguishable, i.e., $\Pr[\text{Game}_1 = 1] = \Pr[\text{Game}_2 = 1]$, and prove it by inspecting their densities.

Let us first estimate the probability of non-aborts in **Game₁**. When selecting the values $\{\Pi(x_i)\}_{0 \leq i \leq \ell}$, we have in total

$$N_1 = \prod_{k=0}^{\ell-1} (2^\lambda - (q - \ell) - k)$$

possibilities. The rest of the evaluations of Π on all other possible inputs can be chosen freely in $(2^\lambda - \ell)!$ ways. Therefore, the probability of aborting is

$$\Pr[\perp \leftarrow \text{Game}_1] = 1 - \frac{N_1 \cdot (2^\lambda - \ell)!}{2^\lambda!}.$$

Furthermore, any outcome $\vec{y} \in (\{0, 1\}^\lambda)^q$ with non-null density has probability $\frac{(2^\lambda - q)!}{2^\lambda!}$.

Let us now turn to **Game₂**. The abort probability can be verified to be identical. Furthermore, the density of each $\vec{y}$ with non-null density is again $\frac{(2^\lambda - q)!}{2^\lambda!}$. It remains to show that $\vec{y}$ has null density in **Game₁** iff it has null density in **Game₂** too, which trivially follows from the equivalence of the corresponding abort conditions.

Combining the bounds, we conclude that $|\Pr[\text{Game}_0 = 1] - \Pr[\text{Game}_2 = 1]| \leq \ell/2^\lambda$. This immediately implies that F is computationally indistinguishable from the evaluations of a uniformly random permutation. Thanks to the popular PRF/PRP switching lemma [17], this is in turn indistinguishable from a uniformly random function. $\square$

4 Applications to Message Authentication Codes

Having established the existence of 1-adaptive weak PRFs and their greater simplicity in comparison with PRFs, we turn to their applications to message authentication, as anticipated in the technical outline. We prove that, when instantiated with the 1-adaptive weak PRFs from the previous section, the Carter–Wegman–Shoup construction is **uf-cma**-secure (Theorem 10) and **ind-cma**-secure (Theorem 11). This allows us to apply the generic transform by Dodis et al. [32] to obtain a **uf-cmva**-secure MAC (Theorem 13). Afterwards, we provide an analysis of the query complexity of MACs from 1-adaptive weak PRFs.

4.1 Message Authentication Codes

A *message authentication code* (MAC) is an algorithm that computes a tag to verify a message's integrity and authenticity.

Definition 4 (Message Authentication Codes). *A message authentication code Π_{MAC} consists of the following three efficient algorithms:*

KGen: $\{0, 1\}^* \rightarrow \mathcal{D}_\lambda$: *The key-generation algorithm outputs a key $k \in \mathcal{D}_\lambda$ depending on λ .*

$\text{Exp}_{\mathcal{A}, \Pi_{\text{MAC}}}^{\text{uf-cma}}(\lambda)$	$\text{Exp}_{\mathcal{A}, \Pi_{\text{MAC}}}^{\text{uf-cmva}}(\lambda)$	$\mathcal{O}_{\text{Mac}}(k, m)$
1 : $\mathcal{Q} \leftarrow \emptyset, \quad k \leftarrow \text{KGen}(1^\lambda)$	1 : $\mathcal{Q} \leftarrow \emptyset, \quad k \leftarrow \text{KGen}(1^\lambda)$	1 : $T \leftarrow \text{Mac}(k, m)$
2 : $(m^*, t^*) \leftarrow \mathcal{A}^{\mathcal{O}_{\text{Mac}}(k, \cdot)}(1^\lambda)$	2 : $(m^*, t^*) \leftarrow \mathcal{A}^{\mathcal{O}_{\text{Mac}}(k, \cdot), \mathcal{O}_{\text{Vrfy}}(k, \cdot, \cdot, \cdot)}(1^\lambda)$	2 : $\mathcal{Q} \cup \{(m, t)\}$
3 : if $(m^* \notin \mathcal{Q})$:	3 : if $(m^* \notin \mathcal{Q})$:	3 : return t
4 : return $\text{Vrfy}(k, m^*, t^*)$	4 : return $\text{Vrfy}(k, m^*, t^*)$	$\mathcal{O}_{\text{Vrfy}}(k, m, t)$
5 : return 0	5 : return 0	1 : return $\text{Vrfy}(k, m, t)$

Figure 5: Experiments for uf-cma security and uf-cmva security.

Mac: $\mathcal{D}_\lambda \times \mathcal{D}_X \times \mathcal{D}_R \rightarrow \mathcal{D}_Y$: The (probabilistic) tag-generation algorithm takes as input a key k and a message $m \in \mathcal{D}_X$. It samples a random value $r \in \mathcal{D}_R$ and returns a tag $t \in \mathcal{D}_Y$.

Vrfy: $\mathcal{D}_\lambda \times \mathcal{D}_X \times \mathcal{D}_Y \rightarrow \{0, 1\}$: The verification algorithm takes as input a key k , a message m , and a tag t . It outputs a bit b , where 1 accepts the input, and 0 rejects it.

For correctness we require $\text{Vrfy}(k, m, \text{Mac}(k, m)) = 1$, for all messages $m \in \mathcal{D}_X$ and for all keys $k \in \mathcal{D}_\lambda$.

The standard security notion here is *unforgeability under adaptive chosen message attacks* **uf-cma**, meaning an adversary cannot forge a new valid message-tag pair even with access to a tag-generation oracle. We note that in the deterministic case, there is no need to include a verification oracle in the security definition. The reason is that in this case, the adversary can predict the output of the verification oracle: for previously tagged messages, the adversary already knows the unique tag; for new messages, finding such a tag is equivalent to breaking security without the oracle [32].

Definition 5 (uf-cma Security). A message authentication code $\Pi_{\text{MAC}} = (\text{KGen}, \text{Mac}, \text{Vrfy})$ is **uf-cma-secure**, if there exists a negligible function negl such that $\Pr[\text{Exp}_{\mathcal{A}, \Pi_{\text{MAC}}}^{\text{uf-cma}}(\lambda) = 1] \leq \text{negl}(\lambda)$ holds for all PPT adversaries $\mathcal{A}$, where the experiment is defined in Figure 5.

In the case of probabilistic MACs, the security definition needs to include the verification oracle as pointed out by Bellare et al. [13]. Thus, the security notion extends to *unforgeability under adaptive chosen message and verification queries attack* **uf-cmva**.

Definition 6 (uf-cmva Security). A message authentication code $\Pi_{\text{MAC}} = (\text{KGen}, \text{Mac}, \text{Vrfy})$ is **uf-cmva-secure**, if there exists a negligible function negl such that $\Pr[\text{Exp}_{\mathcal{A}, \Pi_{\text{MAC}}}^{\text{uf-cmva}}(\lambda) = 1] \leq \text{negl}(\lambda)$ holds for all PPT adversaries $\mathcal{A}$, where the experiment is defined in Figure 5.

Besides unforgeability, another relevant security notion for MACs is *indistinguishability under chosen message attacks* **ind-cma**. Here, the goal of the adversary is not to create a valid forgery, but rather to distinguish which of two chosen challenge messages was used by the tagging oracle.

Definition 7 (ind-cma Security). A message authentication code $\Pi_{\text{MAC}} = (\text{KGen}, \text{Mac}, \text{Vrfy})$ is **ind-cma-secure** if there exists a negligible function negl such that

$$\Pr \left[b = b' \mid \begin{array}{l} (m_0, m_1) \leftarrow \mathcal{A}^{\mathcal{O}_{\text{Mac}}(k, \cdot)}(1^\lambda) \\ b \leftarrow \{0, 1\} \\ t \leftarrow \text{Mac}(k, m_b) \\ b' \leftarrow \mathcal{A}^{\mathcal{O}_{\text{Mac}}(k, \cdot)}(t) \end{array} \right] \leq \frac{1}{2} + \text{negl}(\lambda),$$

holds for all PPT adversaries $\mathcal{A}$, with $k \leftarrow \$ \text{KGen}(1^\lambda)$, where $\mathcal{A}$ is restricted from querying oracle $\mathcal{O}_{\text{Mac}}$ on the challenge messages m_0 and m_1 .

Carter–Wegman–Shoup Paradigm. The idea of the Carter–Wegman–Shoup paradigm is XOR-ing the output of a strongly universal and secret hash function with some form of randomness. In the initial construction, each message has a unique and random one-time key [71]. Contemporary, this key is replaced by the output of a nonce-input, independently keyed pseudorandom function PRF [68, 27]. In the following, we provide a generalized definition of Carter–Wegman–Shoup style MACs that goes beyond bitstrings and XOR, and encompasses arbitrary additive groups:

Construction 6 (Carter–Wegman–Shoup MACs [71]). *Let $(\mathcal{D}_Y, \oplus)$ be a generic additive group. Let $f: \mathcal{D}_\lambda \times \mathcal{D}_R \rightarrow \mathcal{D}_Y$ be a function and $H: \mathcal{D}_\lambda \times \mathcal{D}^* \rightarrow \mathcal{D}_Y$ be a keyed hash function. For a message m and a key pair (k_1, k_2) we define Π_{MAC} as follows:*

Mac: *On input a key pair (k_1, k_2) as well as a message m , the tag algorithm samples a random value r uniformly at random from $\mathcal{D}_R$, computes $t = H(k_1, m) \oplus f(k_2, r)$ and outputs (t, r) .*

Vrfy: *On input a key pair (k_1, k_2) , a message m , as well as a tag-randomness pair (t, r) , the verification algorithm outputs 1 if $t = t'$ with $t' = H(k_1, m) \oplus f(k_2, r)$ and 0 otherwise.*

Further, Construction 6 uses a keyed hash function, which essentially compresses input of arbitrary length in $\mathcal{D}^*$ to an output of fixed length in $\mathcal{D}_Y$. Let $\text{Func}_Y^{\lambda,*} = \{H \mid H: \mathcal{D}_\lambda \times \mathcal{D}^* \rightarrow \mathcal{D}_Y\}$ be a set of hash function. According to the original description (with shared one-time randomness values), $\text{Func}_Y^{\lambda,*}$ is strongly universal if for any $x_1, x_2 \in \mathcal{D}^*$ with $x_1 \neq x_2$ and any $y_1, y_2 \in \mathcal{D}_Y$, we have $\Pr[H(k, x_1) = y_1 \wedge H(k, x_2) = y_2] = 1/|\mathcal{D}_Y|^2$ with a uniform random key $k \leftarrow \$ \mathcal{D}_\lambda$ instantiating the hash function H [71]. Later Victor Shoup proved that the contemporary construction, which we present in Construction 6, under the assumption that f is a PRF, allows for the relaxation of H being a strongly universal function to H being a XOR-universal function [68]. Due to this analysis and relaxation, we call it the Carter–Wegman–Shoup paradigm.

4.2 Proving Carter–Wegman–Shoup Secure with 1-Adaptive Weak PRFs

It is not possible to prove the uf-cma security of the Carter–Wegman–Shoup paradigm assuming f (for $t = H(k_1, m) \oplus f(k_2, r)$) is a weak pseudorandom function (wPRF). Section 5 provides details for this statement. The intuition arises from the uf-cma security game, which, in line 4, requires the computation of Vrfy on an input adaptively chosen by the adversary. Since Vrfy of Carter–Wegman–Shoup essentially recomputes MAC, this requires at least one adaptive query to f . However, this intuition suggests that with resistance to just one additional adaptive query, the proof is possible. We now formalize this intuition.

Lemma 1. *Let $F: \mathcal{D}_\lambda \times \mathcal{D}_X \rightarrow \mathcal{D}_Y$ be a pseudorandom function. Let $\text{pwF}: \mathcal{D}_\lambda \times \mathcal{D}_X \rightarrow \mathcal{D}_Y$ be a 1-adaptive weak pseudorandom function according to Definition 1. Consider the following game $\text{Game}_{1\text{vsFull}}$ involving an interactive adversary $\mathcal{A}$ and parametrized by a function $\ell: \mathbb{N} \rightarrow \mathbb{N}$:*

1. *The challenger flips a random bit $b \leftarrow \$ \{0, 1\}$, and samples two independent keys $k_0, k_1 \leftarrow \$ \mathcal{D}_\lambda$.*

2. The challenger samples $\ell(\lambda)$ random and independent preimages $x_1, \dots, x_{\ell(\lambda)} \leftarrow \$ \mathcal{D}_X$. It returns $\{(x_i, y_i)\}_{i \in [\ell(\lambda)]}$ to $\mathcal{A}$, where

$$y_i = \begin{cases} F(k_0, x_i) & b = 0 \\ \text{pwF}(k_1, x_i) & b = 1 \end{cases}.$$

3. We allow $\mathcal{A}$ to make a single adaptive query with $x^* \in \mathcal{D}_X$. For that, $\mathcal{A}$ has access to an oracle $\mathcal{O}$. If $b = 0$, on input $x \in \mathcal{D}_X$, oracle $\mathcal{O}$ returns $F(k_0, x)$. If $b = 1$, it returns $\text{pwF}(k_1, x)$.
4. $\mathcal{A}$ outputs a guess bit b' . The game's output is $\neg(b \oplus b')$.

Then for any PPT adversary $\mathcal{A}$ and polynomial-bounded function ℓ , there exists a negligible function negl such that the advantage of $\mathcal{A}$ in $\text{Game}_{1\text{vsFull}}$, defined as

$$\text{Adv}_{\mathcal{A}, \ell, F, \text{pwF}}^{\text{Game}_{1\text{vsFull}}} = \left| \Pr[\text{Game}_{1\text{vsFull}}(\mathcal{A}) = 1] - \frac{1}{2} \right|$$

is less than or equal to $\text{negl}(\lambda)$.

Proof. We show that the advantage $\text{Adv}_{\mathcal{A}, \ell, F, \text{pwF}}^{\text{Game}_{1\text{vsFull}}}$ of any PPT adversary $\mathcal{A}$ is negligible using a sequence of games. We consider the following sequence of games:

- **Game₀**: This is $\text{Game}_{1\text{vsFull}}$ with a fixed $b = 0$. More precisely, the challenger samples $k_0 \leftarrow \$ \mathcal{D}_\lambda$ and for $\ell = \ell(\lambda)$ samples $x_1, \dots, x_\ell \leftarrow \$ \mathcal{D}_X$. It returns $\{(x_i, F(k_0, x_i))\}_{i \in [\ell]}$ to $\mathcal{A}$. Then, $\mathcal{A}$ can make one adaptive query x^* and gets the resulting y^* from $y^* \leftarrow F(k_0, x^*)$. Eventually, $\mathcal{A}$ outputs a bit b' . By definition, $\Pr[\text{Game}_0 = 1] = \Pr[\mathcal{A} \text{ outputs } 0 \mid b = 0]$.
- **Game₁**: This game is identical to **Game₀**, but we replace the pseudorandom function F with a truly random function R with the same preimage and image space. More precisely, the challenger samples ℓ random $x_1, \dots, x_\ell \leftarrow \$ \mathcal{D}_X$. It returns $\{(x_i, R(x_i))\}_{i \in [\ell]}$ to $\mathcal{A}$. Then, $\mathcal{A}$ can make one adaptive query x^* and gets the resulting y^* from $y^* \leftarrow R(x^*)$. Eventually, $\mathcal{A}$ outputs a bit b' .
- **Game₀ to Game₁**: We claim that

$$|\Pr[\text{Game}_0 = 1] - \Pr[\text{Game}_1 = 1]| \leq \text{Adv}_{\mathcal{R}}^{\text{PRF}}(\lambda).$$

This follows directly from the pseudorandomness of F . We can construct a reduction $\mathcal{R}$ trying to distinguish F from R . Reduction $\mathcal{R}$ has an oracle $\mathcal{O}$ which is either $F(k, \cdot)$ or a random function $R(\cdot)$. Then, $\mathcal{R}$ samples $x_1, \dots, x_\ell$ and queries $\mathcal{O}$ on each x_i to get y_i . $\mathcal{R}$ gives $\{(x_i, y_i)\}_{i \in [\ell]}$ to $\mathcal{A}$. When $\mathcal{A}$ makes its single adaptive query x^* , reduction $\mathcal{R}$ queries $\mathcal{O}(x^*)$ to get y^* . $\mathcal{R}$ gives y^* to $\mathcal{A}$. Eventually, $\mathcal{A}$ outputs a bit b' and $\mathcal{R}$ outputs the same bit b' . Clearly, $\mathcal{R}$ is efficient since it makes $\ell + 1$ queries, which is polynomial. Its advantage in the PRF game is exactly $|\Pr[\text{Game}_0 = 1] - \Pr[\text{Game}_1 = 1]|$. Therefore, the negligible advantage $\text{Adv}_{\mathcal{R}}^{\text{PRF}}(\lambda)$ against the PRF bounds the difference between both games.

- **Game₂**: This game is identical to **Game₁**, but we replace the truly random function R with the 1-adaptive weak pseudorandom function pwF . More precisely, the challenger samples $k_1 \leftarrow \$ \mathcal{D}_\lambda$ and ℓ random $x_1, \dots, x_\ell \leftarrow \$ \mathcal{D}_X$. It returns $\{(x_i, \text{pwF}(k_1, x_i))\}_{i \in [\ell]}$ to $\mathcal{A}$. Then, $\mathcal{A}$ can make one adaptive query x^* and gets the resulting y^* from $y^* \leftarrow \text{pwF}(k_1, x^*)$. Eventually, $\mathcal{A}$ outputs a bit b' . We observe that this is exactly $\text{Game}_{1\text{vsFull}}$ with a fixed $b = 1$.

- **Game₁ to Game₂:** We claim that

$$|\Pr[\text{Game}_1 = 1] - \Pr[\text{Game}_2 = 1]| \leq \text{Adv}_{\mathcal{A}}^{1\text{-awPRF}}(\lambda).$$

This follows directly from the 1-adaptive weak PRF security of **pwF**. The adversary $\mathcal{A}$ in **Game₁** and **Game₂** is, by definition, an adversary for the 1-awPRF security game (see Definition 1). It receives ℓ non-adaptive samples and makes one adaptive query. Therefore, the negligible advantage $\text{Adv}_{\mathcal{A}}^{1\text{-awPRF}}(\lambda)$ against the 1-awPRF bounds the difference between both games.

By the triangle inequality, we have

$$|\Pr[\text{Game}_0 = 1] - \Pr[\text{Game}_2 = 1]| \leq \text{Adv}_{\mathcal{R}}^{\text{PRF}}(\lambda) + \text{Adv}_{\mathcal{A}}^{1\text{-awPRF}}(\lambda).$$

The total advantage of $\mathcal{A}$ in the original game is

$$\text{Adv}_{\mathcal{A}, \ell, \mathbf{F}, \text{pwF}}^{\text{Game}_{1\text{vsFull}}} = \frac{1}{2} |\Pr[\text{Game}_0 = 1] - \Pr[\text{Game}_2 = 1]| \leq \frac{1}{2} \left(\text{Adv}_{\mathcal{R}}^{\text{PRF}}(\lambda) + \text{Adv}_{\mathcal{A}}^{1\text{-awPRF}}(\lambda) \right).$$

Since the sum of two negligible functions is negligible, the advantage of $\mathcal{A}$ is negligible. $\square$

Unforgeability. Due to Victor Shoup [68], we know that the CW construction is a secure MAC when **F** is a full-fledged PRF and H is XOR-universal. We now leverage Lemma 1 to show that this carries over to 1-adaptive weak PRFs.

Theorem 10. *Consider Construction 6. Let **pwF** be a 1-adaptive weak PRF to instantiate f . Let H be a XOR-universal hash function. Then Construction 6 is a **uf-cma**-secure MAC.*

Proof. We prove this by showing that an efficient adversary having a non-negligible advantage against the **uf-cma** experiment implies an efficient adversary with comparably high advantage in **Game_{1vsFull}**. The idea is that a reduction $\mathcal{R}$ in **Game_{1vsFull}** utilizes a subroutine $\mathcal{A}$ and simulates $\text{Exp}_{\mathcal{A}, \Pi_{\text{MAC}}}^{\text{uf-cma}}$ for $\mathcal{A}$, while using the adaptive query to compute the final verification step. Initially, game **Game_{1vsFull}** samples a bit b (besides the necessary keys) with $b = 0$ indicating the computation of a PRF and $b = 1$ indicating the computation of a 1-waPRF. Then, reduction $\mathcal{R}$ receives an input-output pair set $\{(x_i, y_i)\}_{i \in [\ell]}$ of polynomial size, where $x_i \leftarrow \$ \mathcal{D}_R$ and $y_i \in \mathcal{D}_Y$ are either $y_i = \mathbf{F}(k_0, x_i)$ (case $b=0$) or $y_i = \text{pwF}(k_1, x_i)$ (case $b=1$). To simulate the hash function, $\mathcal{R}$ samples $k'_1 \leftarrow \$ \mathcal{D}_\lambda$ for H . Then $\mathcal{R}$ invokes a PPT adversary $\mathcal{A}$ against Construction 6 as a subroutine. W.l.o.g., we assume that the size of $\{(x_i, y_i)\}_{i \in [\ell]}$ is greater or equal to the number of queries q from $\mathcal{A}$ to its **Mac** oracle. On each query m_j by $\mathcal{A}$ to its **Mac** oracles, reduction $\mathcal{R}$ answers by taking an unused pair (x_j, y_j) , setting $r_j \leftarrow x_j$ and $t_j \leftarrow H(k'_1, m_j) \oplus y_j$, and returning (t_j, r_j) . Eventually, $\mathcal{A}$ outputs a forgery (m^*, t^*, r^*) with fresh m^* . Reduction $\mathcal{R}$ must now verify this forgery. It checks two cases for r^* . If r^* is reused, $\mathcal{R}$ looks up the corresponding y_j it received non-adaptively and sets $y^* := y_j$. If r^* is fresh, $\mathcal{R}$ makes its single adaptive query with r^* to obtain the necessary y^* . In either case, $\mathcal{R}$ now has the value y^* and outputs bit $b' = 1$ iff $t^* = H(k'_1, m^*) \oplus y^*$. Otherwise, $\mathcal{R}$ outputs 0.

We have two cases for the analysis of the advantage of $\mathcal{R}$. Let E_0 be the event that $\mathcal{A}$ succeeds in its forgery in the $b = 0$ case, and similarly let E_1 be the event for the $b = 1$ case. The advantage of $\mathcal{R}$ in **Game_{1vsFull}** is $\text{Adv}_{\mathcal{R}}^{\text{Game}_{1\text{vsFull}}} = 1/2 |\Pr[E_0] - \Pr[E_1]|$. In case $b = 0$, the simulation by $\mathcal{R}$ is identical to the real Carter–Wegman–Shoup paradigm with a PRF. $\Pr[E_0]$ is the probability

that $\mathcal{A}$ produces a valid forgery. By the standard Carter–Wegman–Shoup analysis [68], this success probability is bounded by

$$\Pr[E_0] \leq \text{Adv}_{\mathcal{R}}^{\text{PRF}}(\lambda) + \frac{\binom{q}{2}}{|\mathcal{D}_R|} + \frac{1}{|\mathcal{D}_Y|},$$

where $\text{Adv}_{\mathcal{R}}^{\text{PRF}}$ is the advantage against the PRF, $\binom{q}{2}/|\mathcal{D}_R|$ is the probability of an r -collision, and $1/|\mathcal{D}_Y|$ is the forgery probability from the XOR-universality of H (this bounds the r -reuse case). This entire term is negligible. In case $b = 1$, reduction $\mathcal{R}$ perfectly simulates the construction with the 1-awPRF. The probability for E_1 that $\mathcal{R}$ outputs 1 is exactly the success probability of $\mathcal{A}$ against this scheme. Thus, $\Pr[E_1] = \text{Adv}_{\mathcal{A}, \Pi_{\text{MAC}}}^{\text{uf-cma}}(\lambda)$. Putting things together, we get

$$\begin{aligned} \text{Adv}_{\mathcal{R}}^{\text{Game}_{1\text{vsFull}}}(\lambda) &\geq \frac{1}{2} \left(\text{Adv}_{\mathcal{A}, \Pi_{\text{MAC}}}^{\text{uf-cma}}(\lambda) - \text{Adv}_{\mathcal{R}}^{\text{PRF}}(\lambda) - \frac{\binom{q}{2}}{|\mathcal{D}_R|} - \frac{1}{|\mathcal{D}_Y|} \right) \\ &\Leftrightarrow \text{Adv}_{\mathcal{A}, \Pi_{\text{MAC}}}^{\text{uf-cma}}(\lambda) \leq 2 \cdot \text{Adv}_{\mathcal{R}}^{\text{Game}_{1\text{vsFull}}}(\lambda) + \text{Adv}_{\mathcal{R}}^{\text{PRF}}(\lambda) + \frac{\binom{q}{2}}{|\mathcal{D}_R|} + \frac{1}{|\mathcal{D}_Y|}. \end{aligned}$$

Supposing $\text{Adv}_{\mathcal{A}, \Pi_{\text{MAC}}}^{\text{uf-cma}}$ is non-negligible contradicts either Lemma 1 or the PRF assumption. Thus, $\text{Adv}_{\mathcal{A}, \Pi_{\text{MAC}}}^{\text{uf-cma}}$ must be negligible, which concludes our proof. $\square$

Indistinguishability. In contrast to unforgeability, we can prove the indistinguishability of Construction 6 with the assumption of f being a weak PRF. The intuition is that the **ind-cma** security experiment (see Definition 7) does not need to query f adaptively.

Theorem 11. *Consider Construction 6. Let pwF be a weak PRF to instantiate f . Let H be a XOR-universal hash function. Then Construction 6 is an **ind-cma**-secure MAC.*

Proof. We prove this by reduction from the **ind-cma**-secure of the resulting construction towards the weak PRF property of wF . The idea is that reduction $\mathcal{R}$ in the role of a distinguisher for wF utilizes a subroutine $\mathcal{A}$. The reduction does so by simulating $\text{Exp}_{\mathcal{A}, \Pi_{\text{MAC}}}^{\text{ind-cma}}$ for $\mathcal{A}$, and then outputs a bit directly based on the output bit by $\mathcal{A}$. Initially, $\mathcal{R}$ receives a set of $\ell = \ell(\lambda)$ non-adaptive samples $\{(r_i, y_i)\}_{i \in [\ell]}$, where $r_i \leftarrow \mathcal{D}_R$ and y_i are either $y_i = \text{wF}(k, r_i)$ for a random k (real world) or $y_i = R(r_i)$ for a random function R (random world). $\mathcal{R}$ must output a bit to distinguish these worlds. Let q be the polynomial upper bound on the total number of $\mathcal{A}$'s interactions (queries + challenge). W.l.o.g., we assume $\ell \geq q$. Further, we assume that $\mathcal{A}$ does not query its challenge messages to its **Mac** oracle (i.e., we assume the reduction implicitly checks that and aborts in such a case). $\mathcal{R}$ simulates $\text{Exp}_{\mathcal{A}, \Pi_{\text{MAC}}}^{\text{ind-cma}}$ for $\mathcal{A}$ as follows: $\mathcal{R}$ samples $k'_1 \leftarrow \mathcal{D}_\lambda$ for H , initializes a counter i with 1 and then invokes $\mathcal{A}$. Then, $\mathcal{R}$ invokes $\mathcal{A}$. On each query m_j by $\mathcal{A}$, reduction $\mathcal{R}$ takes the i -th pair (r_i, y_i) from T . The reduction computes $t_j \leftarrow H(k'_1, m_j) \oplus y_i$, returns (t_j, r_i) to $\mathcal{A}$, and increments $i \leftarrow i + 1$. Eventually, $\mathcal{A}$ queries a challenge (m_0, m_1) . Then, $\mathcal{R}$ samples its own challenge bit $b \leftarrow \{0, 1\}$, takes the next pair (r_i, y_i) from T , increments i , computes $t_c \leftarrow H(k'_1, m_b) \oplus y_i$, and returns (t_c, r_i) to $\mathcal{A}$. Afterwards, $\mathcal{R}$ continues answering further queries by $\mathcal{A}$ in the same manner as described above. Eventually, $\mathcal{A}$ outputs its guess b' . $\mathcal{R}$ outputs 1 if $b = b'$, and 0 otherwise.

We have two cases for the analysis of the advantage of $\mathcal{R}$. Let E_{real} be the event that $\mathcal{R}$ outputs 1 in the real world case, and similarly let E_{rand} be the event for the random world case. The advantage

of $\mathcal{R}$ is $\text{Adv}_{\mathcal{R}}^{\text{wPRF}} = |\Pr[E_{\text{real}}] - \Pr[E_{\text{rand}}]|$. In the real world case, the simulation is perfect. The y_j values are $y_j = \text{wF}(k, r_j)$. The view of $\mathcal{A}$ is identical to the ind-cma experiment, which gives us

$$\Pr[E_{\text{real}}] = \frac{1}{2} + \text{Adv}_{\mathcal{A}}^{\text{ind-cma}}(\lambda).$$

In the random world case, the y_j values are from R . Let c be the index for the challenge. The challenge value is $y_c = R(r_c)$. Let E_{coll} be the event of collision $r_c = r_j$ for some $j \neq c$. We use the common bound $\Pr[E_{\text{coll}}] \leq (q-1)/|\mathcal{D}_R|$. If $\neg E_{\text{coll}}$ holds, y_c is uniform and independent of all other y_j ($j \neq c$) and k'_1 . The challenge $t_c = H(k'_1, m_b) \oplus y_c$ is a one-time pad, and $\mathcal{A}$'s view is statistically independent of b , which gives us

$$\Pr[E_{\text{rand}}] \leq \frac{1}{2} + \Pr[E_{\text{coll}}] \leq \frac{1}{2} + \frac{q-1}{|\mathcal{D}_R|}.$$

Putting things together, we get

$$\text{Adv}_{\mathcal{R}}^{\text{wPRF}}(\lambda) = \text{Adv}_{\mathcal{A}}^{\text{ind-cma}}(\lambda) - \frac{q-1}{|\mathcal{D}_R|} \Leftrightarrow \text{Adv}_{\mathcal{A}}^{\text{ind-cma}}(\lambda) \leq \text{Adv}_{\mathcal{R}}^{\text{wPRF}}(\lambda) + \frac{q-1}{|\mathcal{D}_R|}.$$

Supposing $\text{Adv}_{\mathcal{A}}^{\text{ind-cma}}$ is non-negligible contradicts $\text{Adv}_{\mathcal{R}}^{\text{wPRF}}$ being negligible. Thus, $\text{Adv}_{\mathcal{A}}^{\text{ind-cma}}$ must be negligible, which concludes our proof. $\square$

4.3 Black-box uf-cmva MAC Construction from 1-Adaptive Weak PRFs

For a comprehensive presentation of the necessary material, we now introduce a generic transformation by Dodis et al. [32], which provides a uf-cmva secure scheme Π_{MAC}^t from any uf-cma and ind-cma secure Π_{MAC} scheme. This corresponds to Theorem 3.4 in their paper. We present a simplified write-up in our notation of the transform in Construction 7 with corresponding Theorem 12. We use this transform to lift the Carter–Wegman–Shoup paradigm under the 1-awPRF assumption from uf-cma and ind-cma security to uf-cmva , which is the standard security guarantee of probabilistic MAC schemes. We formally state this in Theorem 13.

Construction 7 (Generic MAC Transform [32]). *Let $\Pi_{\text{MAC}} = \{\text{KGen}, \text{Mac}, \text{Vrfy}\}$ be a MAC scheme with key space $\mathcal{D}_{\lambda}^m$, message space $\mathcal{D}_X^m \times \mathcal{D}_Y^h$, and tag space $\mathcal{D}_Y^m$. Let $H: \mathcal{D}_{\lambda}^h \times \mathcal{D}_Y^m \rightarrow \mathcal{D}_Y^h$ be a (keyed) hash function. Then, the construction of $\Pi_{\text{MAC}}^t = \{\text{KGen}_t, \text{Mac}_t, \text{Vrfy}_t\}$ with key space $\mathcal{D}_{\lambda}^m \times \mathcal{D}_{\lambda}^h$, message space $\mathcal{D}_X^m$, and tag space $\mathcal{D}_Y^m \times \mathcal{D}_Y^h$ is as follows:*

KGen_t: *This key-generation algorithm runs $k_1 \leftarrow \text{KGen}(1^{\lambda})$, uniformly randomly samples a key k_2 from $\mathcal{D}_{\lambda}^h$ to instantiate the hash function, and outputs (k_1, k_2) as the secret key.*

Mac_t: *On input a key tuple (k_1, k_2) as well as a message m , the tag algorithm samples a randomness b uniformly at random from $\mathcal{D}_Y^h$, computes tag $t = \text{Mac}(k_1, m || b)$, and outputs $(t, H(k_2, t) \oplus b)$.*

Vrfy_t: *On input a key tuple (k_1, k_2) , a message m , as well as a pair (t, π) , the verification algorithm computes $b' = H(k_2, t) \oplus \pi$ and outputs $\text{Vrfy}(k_1, m || b', t)$.*

The corresponding (simplified) theorem by Dodis et al. [32] is as follows.

Theorem 12 ($\text{uf-cma} + \text{ind-cma} \Rightarrow \text{uf-cmva}$ [32]). *Consider Π_{MAC} and Π_{MAC}^t from Construction 7. If Π_{MAC} is uf-cma secure and ind-cma secure, then Π_{MAC}^t is uf-cmva secure.*

Due to Theorem 10 and Theorem 11, we know that Carter–Wegman–Shoup scheme $\Pi_{\text{MAC}}^{\text{CWS}}$ is **uf-cma** and **ind-cma** secure under a 1-adaptive weak PRF assumption. Consequently, it directly follows from the results of Dodis et al. [32] that applying the generic transform from Construction 7 to $\Pi_{\text{MAC}}^{\text{CWS}}$ gives a **uf-cmva** secure MAC, which we now formally state and prove.

Theorem 13 (MAC from awPRFs). *Let $\Pi_{\text{MAC}}^{\text{CWS}}$ be the Carter–Wegman–Shoup scheme from Construction 6, instantiated with a 1-adaptive weak PRF for f and a XOR-universal hash function for H . Let Π_{MAC}^t be the result of applying Construction 7 to $\Pi_{\text{MAC}}^{\text{CWS}}$. Let the transform’s hash function, denoted as H' , be XOR-universal. Then, Π_{MAC}^t is a **uf-cmva**-secure MAC.*

Proof. The proof is a direct application of the security guarantee from Theorem 12. We instantiate the underlying scheme Π_{MAC} with our Carter–Wegman–Shoup scheme, $\Pi_{\text{MAC}}^{\text{CWS}}$. Theorem 12 requires $\Pi_{\text{MAC}}^{\text{CWS}}$ to be both **uf-cma** and **ind-cma** secure. By Theorem 10, $\Pi_{\text{MAC}}^{\text{CWS}}$ is **uf-cma**-secure. By Theorem 11, $\Pi_{\text{MAC}}^{\text{CWS}}$ is also **ind-cma**-secure. We note that any 1-awPRF is also a wPRF. Theorem 12 states that the transform yields a **uf-cmva**-secure scheme and therefore Π_{MAC}^t is **uf-cmva**-secure. $\square$

5 (In)-Feasibility of Carter–Wegman–Shoup From Weak PRFs

Many MAC schemes, including the Carter–Wegman–Shoup paradigm (see Construction 6), base their security on a full PRF. We recall this paradigm computes a tag in **Mac** through a PRF f that is evaluated on a uniformly random nonce r independent of the message. The intuition that modelling f as a PRF utilizes a stronger than necessary assumption motivated the definition of 1-adaptive weak PRFs, as well as the associated constructions and proofs. This is because r has high entropy, while under the PRF assumption, the evaluation of $f(k, x)$ is pseudorandom for any preimage x , regardless of its entropy.

Because this intuition was successful once, we wonder whether one can go one step further and leverage even less structured cryptographic objects that exploit some of the input’s entropy. Specifically, weak PRFs appear as a natural candidate replacement. Due to the generic transform by Dodis et al. [32] giving a **uf-cmva**-secure MAC from a **uf-cma** and **ind-cma** MAC scheme, we can restrict our analysis to the latter two security notions. Theorem 11 shows that weak PRFs suffice for **ind-cma** security, thus leaving **uf-cma** as the last remaining property to inspect. Unfortunately, we can prove that weak PRFs are insufficient to prove Carter–Wegman–Shoup **uf-cma**-secure. While at first sight f is only evaluated on uniformly random nonces, there is a hidden adaptive query at the end of $\text{Exp}_{\mathcal{A}, \Pi_{\text{MAC}}}^{\text{uf-cma}}$ (see Definition 5): the challenger must compute **Vrfy** in line 4, which in turn requires evaluating f on an adversarially provided (and so, not necessarily random) input. Although this may seem like a minor technical obstacle, this verification query has substantive implications. More precisely, we show that no conceivable security reduction in which f is modeled as a wPRF and H is a XOR-universal hash is possible. The result can be further strengthened to a version where H is strongly universal.

Before diving into the claimed results, we highlight the intuitions at their core. The first two impossibility claims follow from the fact that a weak PRF could be perfectly predictable on a fixed input $\bar{x} \in \mathcal{D}_X$ without violating its definition. When the inputs to the weak PRF are sampled at random, $\bar{x}$ occurs with negligible probability; but given the possibility of an adaptive query, $\mathcal{A}$ can always cause the weak PRF to be evaluated over $\bar{x}$. As we will see, this mechanism can be exploited for forging.

The second set of impossibility results stems from a particular variant of wPRFs that, namely, the approximately input-homomorphic ones. Informally, if $(\mathbb{G}, +)$ and $(\mathbb{G}', +')$ denote two finite additive groups, and $\mathbf{wF}$ is a weak PRF with $\mathcal{D}_X = \mathbb{G}$ and $\mathcal{D}_Y = \mathbb{G}'$, $\mathbf{wF}$ is approximately input-homomorphic if for all $x, y \in \mathbb{G}$

$$\mathbf{wF}(k, x) +' \mathbf{wF}(k, y) = \mathbf{wF}(k, x + y) + e(x, y),$$

and there exists a PPT algorithm G that guesses $e(x, y)$ with non-negligible probability. Many standard wPRFs satisfy this notion. For instance, the popular DDH-based PRF $\mathbf{F}(k, x) = kx$ is perfectly input-homomorphic, since by construction $\mathbf{wF}(k, x) + \mathbf{wF}(k, y) = kx + ky = \mathbf{wF}(k, x + y)$. On the other hand, the LWR-based weak PRF $\mathbf{wF}(k, x) = \lfloor \langle k, x \rangle \rfloor_p$ with $k, z \in \mathbb{Z}_q^\ell$ and $p < q$, enjoys the property that for any two inputs x and y it holds

$$\mathbf{wF}(k, x) + \mathbf{wF}(k, y) = \lfloor \langle k, x \rangle \rfloor_p + \lfloor \langle k, y \rangle \rfloor_p = \lfloor \langle k, x + y \rangle \rfloor_p - e,$$

and e can be shown to be either 0 or 1. Therefore, one can guess it at random with probability $1/2$.

Additive homomorphisms are a peculiarity of weak PRFs, which cannot arise in full PRFs³. As we will shortly see, this property can be leveraged to combine previously seen tags into a new tag for a fresh message, namely through a simple linear combination.

We now present our impossibility results. The first two hold *unconditionally*, in the sense that they are not based on any specific computational assumption.

Theorem 14. *Consider Construction 6 instantiated over $\{0, 1\}^\lambda$. Then there exists no black-box reduction from uf-cma security to the weak pseudorandomness of f if H is XOR-universal.*

Proof. We prove the thesis by exhibiting a counterexample, i.e., a pair $(H, \mathbf{wF})$ such that H is XOR-universal, $\mathbf{wF}$ is a weak PRF, but their combination according to Construction 6 is not uf-cma-secure. Let $\mathbf{F}: \mathcal{D}_\lambda \times \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ be a PRF. Define a new keyed function $\mathbf{wF}: \mathcal{D}_\lambda \times \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ such that

$$\mathbf{wF}(k, x) = \begin{cases} \mathbf{F}(k, x) & x \neq 0 \\ 0 & x = 0 \end{cases}.$$

$\mathbf{wF}$ is a weak PRF, since for any $q = \text{poly}(\lambda)$ number of randomly sampled inputs, the probability of sampling 0 is at most $q/2^\lambda$. Let H be the keyed hash function $H(k, x) = k \cdot x$, where $k \in \mathbb{F}_{2^\lambda}$, and x is implicitly interpreted as an element of the same finite field. It is straightforward to verify that H is a XOR-universal hash function. If one instantiates Construction 6 with the aforementioned building blocks, $\mathcal{A}$ can immediately output the forgery $(m^*, t^*) = (0, (0, 0))$ without any tagging query. The latter triggers a victory because $H(k, 0) \oplus \mathbf{wF}(k', 0) = 0$ for any k, k' . $\square$

We now generalize our result. We show similar results for a stricter notion of universality for the hash function, namely strong universality (or pairwise independence) instead of XOR-universality. Further, we lift the notion of security to strong unforgeability **suf-cma**.

Essentially, **suf-cma** is a straightforward adaptation of the **uf-cma** security experiment in Definition 5. The difference is that in the experiment formalization (see Figure 5) line 3 is replaced by **if** $((m^*, t^*) \notin \mathcal{Q})$.

³Otherwise, a distinguisher D could simply query the PRF on inputs $x, y, x + y$ and use the guessing algorithm to predict the evaluation at point $x + y$

Theorem 15. *Consider Construction 6 instantiated over $\{0, 1\}^\lambda$. Then there exists no black-box reduction from suf-cma security to the weak pseudorandomness of f if H is strongly universal.*

Proof. We once again prove the claim through a counterexample. Let $F: \mathcal{D}_\lambda \times \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ be a PRF. Define a new keyed function $\mathbf{wF}: \mathcal{D}_\lambda \times \{0, 1\}^\lambda \rightarrow \{0, 1\}^\lambda$ such that

$$\mathbf{wF}(k, x) = \begin{cases} F(k, x) & x \equiv 0 \pmod{2} \\ F(k, x - 1) & x \equiv 1 \pmod{2} \end{cases}.$$

In the above definition, we implicitly convert x to an integer. We now prove that $\mathbf{wF}$ is a weak PRF. To do so, let $y(x)$ be the input to F , i.e., $y(x) = x$ if x is even and $y(x) = x - 1$ if x is odd. This implies $\mathbf{wF}(k, x) = F(k, y(x))$. Consider $q = \text{poly}(\lambda)(\lambda)$ randomly sampled inputs $x_1, \dots, x_q$. A collision in the inputs to F occurs if $y(x_i) = y(x_j)$ for some $i \neq j$. This happens iff $\{x_i, x_j\} = \{z, z-1\}$ for some odd integer z . We call this event E . For any pair (x_i, x_j) with $i \neq j$, the probability E is $\Pr[y(x_i) = y(x_j) \wedge x_i \neq x_j] = 2^{-\lambda}$. By using a union bound we get $\Pr[E] \leq \binom{q}{2} 2^{-\lambda}$, which is negligible for any polynomial q . In the case of $\neg E$, all q values $y(x_1), \dots, y(x_q)$ are distinct. Due to the pseudorandomness of F , the output tuple $(F(k, y(x_1)), \dots, F(k, y(x_q)))$ is computationally indistinguishable from a tuple of q independent uniform random values. Since the advantage of any weak PRF distinguisher against $\mathbf{wF}$ is bounded by the PRF advantage against F plus the negligible probability $\Pr[E]$, we conclude that $\mathbf{wF}$ is a weak PRF.

Let H denote any strongly universal hash function, and $\vec{k}$ the joint key of H and $\mathbf{wF}$. We can now detail the following attack against that breaks suf-cma :

1. $\mathcal{A}$ queries $\text{Mac}(\vec{k}, 0)$, obtaining a tag (t, r) .
2. Convert r into an integer. Output a forgery $(m^*, t^*) = (0, (t, r + (-1)^{r \bmod 2}))$.

The forgery is valid because $\mathbf{wF}(k, x) = \mathbf{wF}(k, x + (-1)^{x \bmod 2})$ for all k by construction, and so $t = H(k, 0) \oplus \mathbf{wF}(k', r) = H(k, 0) \oplus \mathbf{wF}(k', r + (-1)^{r \bmod 2})$. $\square$

Theorems 14 and 15 suffice to prove that weak PRFs are unsuitable for MACs with building blocks instantiated over $\{0, 1\}^\lambda$. In the following, we show that the impossibility is broader and targets a more general algebraic structure. Concretely, we exhibit two impossibility results, targeting instantiations of Construction 6 over $(\mathbb{Z}_p, +)$ with prime p , and $(\mathbb{G}, +)$, with the latter representing an additive group assuming DDH hard. Both impossibility results depend on two well-established computational assumptions, namely LWE (with appropriate parameters) and DDH .

Theorem 16. *Consider Construction 6 instantiated over $\mathbb{Z}_p$ with prime p . Then there exists no black-box reduction from uf-cma security to the weak pseudorandomness of f if H is strongly universal and the LWR assumption with parameters (q, p, m, n) with $q < p$ and arbitrary $m, n \in \mathbb{N}$ is true.*

Proof. Let $\mathbf{wF}$ denote the standard LWR -based weak PRF (see Section C.2 for the extended definition). Let H denote the standard strongly universal hash over finite prime fields $H(a, b, x) = ax + b$, with key $(a, b) \in \mathbb{Z}_p^2$ (see Section C.1). Let $\vec{k} = (k, a, b)$, with k denoting the key of $\mathbf{wF}$. An adversary $\mathcal{A}$ can break the uf-cma security of Construction 6 instantiated with the aforementioned building blocks as follows:

1. Pick two arbitrary distinct messages $m_1, m_2 \in \mathbb{Z}_p$.

2. Obtain $(t_1, r_1) \leftarrow \text{Mac}(\vec{k}, m_1)$.
3. Obtain $(t_2, r_2) \leftarrow \text{Mac}(\vec{k}, m_2)$.
4. Obtain $(t_0, r_0) \leftarrow \text{Mac}(\vec{k}, 0)$.
5. Sample random $e \leftarrow_{\$} \{-1, 0, 1\}$.

Notice that by construction $\text{wF}(k', r_1) + \text{wF}(k', r_2) - \text{wF}(k', r_0) = \text{wF}(k', r_1 + r_2 - r_0) - \epsilon$, with $\epsilon \in \{-1, 0, 1\}$. In the attack, $\Pr[e = \epsilon] = 1/3$. Furthermore, $H(a, b, m_1) + H(a, b, m_2) - H(a, b, 0) = H(a, b, m_1 + m_2)$. Therefore, conditioned on correctly guessing ϵ , we have

$$t_1 + t_2 - t_0 + e = t_1 + t_2 - t_0 + \epsilon = H(a, b, m_1 + m_2) + \text{wF}(k', r_1 + r_2 - r_0) + 0.$$

This implies that output forgery is valid, despite $\mathcal{A}$ never querying MAC on input $m_1 + m_2$. The success probability is exactly $1/3$, which concludes our proof. $\square$

Theorem 17. *Consider Construction 6 instantiated over a finite cyclic additive group $(\mathbb{G}, +)$ of prime order p . Then there exists no black-box reduction from uf-cma security to the weak pseudo-randomness of f if H is strongly universal and the DDH assumption is true in $\mathbb{G}$.*

Proof. Let wF denote the DDH-based weak PRF defined as $\text{wF}(k, x) = kx$ with $k \in \mathbb{Z}_p$ and $x \in \mathbb{G}$ (see Section C.2). We introduce a strongly universal hash function H , which reads a key $a, b \in \mathbb{Z}_p \times \mathbb{G}$, an input $x \in \mathbb{G}$, and outputs $H(a, b, x) = ax + b$ (see Section C.1 for a proof of strong universality). Let $\vec{k} = (k, a, v)$. We now show how an adversary $\mathcal{A}$ can output a forgery:

1. Pick two arbitrary distinct messages $m_1, m_2 \in \mathbb{Z}_p$.
2. Obtain $(t_1, r_1) \leftarrow \text{Mac}(\vec{k}, m_1)$.
3. Obtain $(t_2, r_2) \leftarrow \text{Mac}(\vec{k}, m_2)$.
4. Obtain $(t_0, r_0) \leftarrow \text{Mac}(\vec{k}, 0)$.
5. Output $(m^*, t^*) = (m_1 + m_2, (t_1 + t_2 - t_0 - e, r_1 + r_2 - r_0))$.

The forgery is valid by a reasoning akin to that of Theorem 16, but without considering the approximation error ϵ . $\square$

The previous theorems do not, in principle, rule out the possibility of applying some form of pre-processing or postprocessing to the weak PRF to enable a security reduction. However, we point out that by minimal tweaks we can rule out some natural classes of pre- and post-processing strategies. In the following, we provide a non-exhaustive list of the adaptations and their implications:

- **Adapting Theorem 14:** the theorem can be easily adapted to exclude black-box security reductions for instantiations of Construction 6 where $\mathbf{F}$ is of the form $\text{wF}(k, \tilde{H}(\cdot))$ or $\tilde{H}(\text{wF}(k, \cdot))$, with wF a weak PRF, and $H, \tilde{H}$ XOR-universal hashes. We can recycle the counterexample of Theorem 14 by keeping H and wF unchanged, and setting $\tilde{H}$ to equal H , but with an independently sampled key $\tilde{k}$. Then, we would still get

$$H_k(0) \oplus \tilde{H}_{\tilde{k}}(\text{wF}(k', 0)) = H_k(0) \oplus \text{wF}(k', \tilde{H}_{\tilde{k}}(0)) = 0.$$

- **Adapting Theorem 16:** similarly, we can adapt the counterexample of Theorem 16 to rule out preprocessing of the form $\mathbf{wF}(k, \tilde{H}(\cdot))$ or $\tilde{H}(\mathbf{wF}(k, \cdot))$, with $\mathbf{wF}$ a weak PRF, and $H, \tilde{H}$ strongly universal hashes. Again, we recycle the counterexample by keeping H and $\mathbf{wF}$ unchanged, and setting $\tilde{H}$ to be instantiated as H , but with an independently sampled key $\tilde{k} = (\tilde{a}, \tilde{b})$. A straightforward generalization of the previous attack yields

$$\begin{aligned} & \left(H_{a,b}(m_1) + \tilde{H}_{\tilde{a},\tilde{b}}(\mathbf{wF}(k', r_1)) \right) + \left(H_{a,b}(m_2) + \tilde{H}_{\tilde{a},\tilde{b}}(\mathbf{wF}(k', r_2)) \right) - \left(H_{a,b}(0) + \tilde{H}_{\tilde{a},\tilde{b}}(\mathbf{wF}(k', r_0)) \right) \\ &= H_{a,b}(m_1 + m_2) + \tilde{H}_{\tilde{a},\tilde{b}}(\mathbf{wF}(k', r_1 + r_2 - r_0)). \end{aligned}$$

Furthermore, it is straightforward to extend the above to the preprocessing of type $\mathbf{wF}(k, \tilde{H}(\cdot))$.

- **Adapting Theorem 17:** through the same reasoning, we can adapt Theorem 17 to rule out the same type of pre- and post-processing.

6 Applications: DDH and LWR MACs

In this section, we focus on concrete instantiations of our construction of 1-adaptive weak PRFs based on standard cryptographic assumptions.

6.1 Bounded PRF C .

Our construction above requires a $\log(\lambda)^2$ -bounded pseudorandom function C mapping $\lambda - \log(\lambda)$ bits to λ bits, which we instantiate as a $\log(\lambda)^2$ -wise independent function. Due to inhomogeneities between the input and output size, we view its key as the coefficients of two polynomials $f, g \in \mathbb{F}_Q[X]$ of degree $d = \log(\lambda)^2$ with $Q = 2^{\lambda - \log(\lambda)}$. When evaluating C on inputs $x \in \mathbb{F}_Q$, we compute $Y_f \leftarrow f(x)$, $Y_g \leftarrow g(x)$, and output the λ most significant bits of the binary string $Y_f || Y_g$ (where the finite field elements are implicitly converted into bitstrings beforehand).

C has a key size of $2d \log Q = 2 \log(\lambda)^2 (\lambda - \log(\lambda)) = \mathcal{O}(\log(\lambda)^2 \lambda)$. Note that we do not have to store it, as it is computed at runtime through a small-domain PRF. However, this parameter remains relevant, as it determines the runtime complexity of C .

6.2 A DDH-Based Instantiation

Setup. We assume a cyclic group $\mathbb{G}$ of size p and generator g in which the DDH problem is hard. Furthermore, we assume p has bitlength λ .

Small-domain PRFs. We instantiate two small-domain PRFs S and S' , with domain size $\log(\lambda)^2$ and $\log(\lambda)$ each, through the standard DDH-based Naor-Reingold pseudorandom function [58], as described in Section C.3. S' is used to generate the key of C , i.e., the $2d$ elements of $\mathbb{F}_Q$. The secret key of S' consists of $\vec{k} \leftarrow_{\$} \mathbb{Z}_p^{(1+\log \lambda) \times 2d}$. At first, on input $x \in \{0, 1\}^{\log \lambda}$ we define S'_{lg} as

$$S'_{lg}(\vec{k}, x) = \begin{pmatrix} g^{k_{1,0} \prod_{i \in [\log \lambda]} k_{1,i}^{x_i}} \\ \vdots \\ g^{k_{2d,0} \prod_{i \in [\log \lambda]} k_{2d,i}^{x_i}} \end{pmatrix}.$$

The total key size is $2d(\log \lambda + 1) \log p = \mathcal{O}(\log(\lambda)^3 \lambda)$. The output of S'_{lg} consists of $2d$ elements $g_1, \dots, g_{2d} \in \mathbb{G}$. Each of them contains λ bits of entropy, out of which we can extract the desired $\log Q$ pseudorandom bits through a deterministic extractor Ext . Therefore, we define S' as

$$S'(\vec{k}, x) = \text{Ext}\left(S'_{lg}(\vec{k}, x)\right).$$

S must instead map $\{0, 1\}^{\log(\lambda)^2}$ into the key space of S' , namely $\mathbb{Z}_p^{(1+\log \lambda) \times 2d}$. We can once again instantiate it through the Naor-Reingold PRF. S takes a key $\vec{k} \leftarrow \mathbb{Z}_p^{\log(\lambda)^2 \times 2d(1+\log \lambda)}$, and outputs on input x

$$S(\vec{k}, x) = \begin{pmatrix} g^{k_{1,0} \prod_{i \in [\log(\lambda)^2]} k_{1,i}^{x_i}} \\ \vdots \\ g^{k_{2d(1+\log \lambda),0} \prod_{i \in [\log(\lambda)^2]} k_{2d(1+\log \lambda),i}^{x_i}} \end{pmatrix}.$$

Comparison with PRFs. We now compare our DDH-based instantiation with its counterpart, the DDH-based Naor-Reingold PRF. We summarize the results of this discussion in Figure 6. The key used in the last step represents the only key material that we effectively need to store. Therefore, our total key size is $\mathcal{O}(\log(\lambda)^5 \lambda)$. On the other hand, directly using the Naor-Reingold PRF to process inputs of size λ requires keys of size $\mathcal{O}(\lambda^2)$. From the viewpoint of time complexity, instead, the full-fledged Naor-Reingold PRF requires λ multiplications in $\mathbb{Z}_p$. The total cost of our solution amounts to the following:

- $2d(1 + \log \lambda) \log(\lambda)^2 = \mathcal{O}(\log(\lambda)^5)$ multiplications modulo p to evaluate S .
- $2d \log \lambda = \mathcal{O}(\log(\lambda)^3)$ multiplications modulo p to evaluate S_{lg} , which when combined with the complexity of the randomness extractor yields the total cost of S' .
- The extractor is heavily group-dependent, but for reasonable choices of $\mathbb{G}$ its asymptotic cost is $\mathcal{O}(1)$. For instance, if $\mathbb{G}$ denotes the group of square residues modulo a safe prime $q = 2p + 1$, Ext can just convert to an element of $\mathbb{Z}_p$, and extract the first $\log Q$ bits. If one instantiates $\mathbb{G}$ as an elliptic curve group over a quadratic field, constant-time extractors were previously considered both in theory [34] and in applications [61].
- $d = \mathcal{O}(\log(\lambda)^2)$ multiplications and an equal number of additions in $\mathbb{F}_Q$ for C .

Let $M(\kappa)$ and $A(\kappa)$ denote the complexity of multiplication and addition in $\mathbb{F}_\kappa$, respectively. Then our total is as follows:

- $\mathcal{O}(\lambda \cdot M(p))$ elementary operations for a PRF.
- $\mathcal{O}(\log(\lambda)^5 M(p) + \log(\lambda)^2 M(Q) + \log(\lambda)^2 A(Q))$ for our instantiation.

In practice, one should choose a multiplication algorithm from the pool of currently available ones. For the sake of simplicity, we assume a multiplication algorithm of complexity $\mathcal{O}(\lambda \log \lambda)$, as achieved by [42], which asymptotically upper bounds both $M(p)$ and $M(Q)$. Similarly, we get a complexity for addition of $\mathcal{O}(\lambda)$ in both cases. Combining everything, we go from a complexity of $\mathcal{O}(\lambda^2 \log \lambda)$ to $\mathcal{O}(\log(\lambda)^5 \lambda)$.

6.3 A Lattice-Based Instantiation

Small-domain PRFs. We instantiate two small-domain PRFs S and S' , with domain size $\log(\lambda)^2$ and $\log(\lambda)$ each, through the standard BLMR pseudorandom function [26], whose details are provided in Section C.3. Note that, by construction, the output of S' should be the key used by C , with the latter described above. Therefore, it should map $\log(\lambda)$ input bits to $2d$ elements of $\mathbb{F}_Q$. For this purpose, we sample a key $\vec{k} \leftarrow \$\mathbb{Z}_q^m$ for an integer m to be fine-tuned shortly and two moduli q_0, q_1 such that $q_0 \mid q_1$. We also assume that the two public matrices $A_0, A_1 \in \mathbb{Z}_{q_1}^{m \times m}$ required by the BLMR PRF were appropriately generated beforehand. On input a bitstring $x \in \{0, 1\}^{\log \lambda}$ we define S' as

$$S'(\vec{k}, x) = \left[\prod_{i \in [\log \lambda]} A_{x_i} \cdot \vec{k} \right]_{q_0}.$$

S must map bitstrings of length $\log(\lambda)^2$ to the keyspace of S' , which we have established to be $\mathbb{Z}_{q_1}^m$. We then sample a key $\vec{k}' \leftarrow \$\mathbb{Z}_{q_2}^m$ for a modulus q_2 such that $q_1 \mid q_2$, assume two suitably sampled public matrices $B_0, B_1 \in \mathbb{Z}_{q_1}^{m \times m}$, and define

$$S(\vec{k}, x) = \left[\prod_{i \in [\log(\lambda)^2]} B_{x_i} \cdot \vec{k}' \right]_{q_1}.$$

We must still tune the relevant parameters (i.e., m, q_0, q_1, q_2) to ensure compliance with the specifications of Construction 3. Let us exhibit a possible strategy. S' outputs a total of $m \log q_0$ bits, and C instead requires $2d \log Q$ bits. We further recall that by Theorem 18, it must hold $m = \lceil n \log q_1 \rceil$, and $n = \lambda$ for us. We get to an inequality⁴

$$2d \log Q = 2 \log(\lambda)^2 (\lambda - \log \lambda) \leq \lceil \lambda \log q_1 \rceil \log q_0 = m \log q_0.$$

The above is easily met for q_0 and q_1 , both polynomial in λ . On top of this, we should tune parameters to ensure that a certain instance of LWE, from which BLMR inherits its hardness, is computationally intractable. To this end, we follow the blueprint proposed by the authors themselves [26]:

1. Make the ansatz $q_1 = \mathcal{O}(\sqrt{\lambda}/\alpha)$, where $\alpha \in (0, 1)$ is a parameter tuning the hardness of LWE in Theorem 18.
2. For a constant $c > 0$, set $\alpha = 2^{-\omega(\log(\lambda)^{1+c})}$, leading to $q_1 = \mathcal{O}(\sqrt{\lambda} 2^{\omega(\log(\lambda)^{1+c})})$.
3. The authors prove that this setup supports inputs of length $\mathcal{O}(\log \lambda)$, as required by S' , as long as $q_0 = 2^{\log(\lambda)^{1+c}}$.

This strategy is advantageous for us, as it leads to

$$\lceil \lambda \log q_1 \rceil \log q_0 \geq \lambda \cdot (\log(\lambda)/2 + \omega(\log(\lambda)^{1+c})) \cdot \log(\lambda)^{1+c}$$

⁴If the output length slightly exceeds $2d \log Q$, one can just discard the excess random bits.

	Naor-Reingold PRF	DDH 1-adaptive weak PRF	BLMR PRF	LWR 1-adaptive weak PRF
Assumption	DDH	DDH	LWE	LWE
Space	$\mathcal{O}(\lambda^2)$	$\mathcal{O}(\lambda \log(\lambda)^5)$	-	$\mathcal{O}(m \log q_2)$
Time(Param.)	$\mathcal{O}(\lambda M(p))$	$\mathcal{O}(\log(\lambda)^5 M(p) + \log(\lambda)^2 M(Q) + \log(\lambda)^2 A(Q))$	-	$\mathcal{O}(\log(\lambda)^{2+2\omega} M(q_2) + \log(\lambda)^2 M(Q))$
Time(Inst.)	$\mathcal{O}(\lambda^2 \log \lambda)$	$\mathcal{O}(\lambda \log(\lambda)^5)$	-	-
Notes	-	-	Not secure for linear input length	-

Figure 6: Comparison between our instantiations of Construction 1 and full-fledged PRFs based on analogous assumptions. $M(\kappa)$ and $A(\kappa)$ denote, for any prime power κ , the computational complexity of multiplication and addition over $\mathbb{F}_\kappa$, respectively. Similarly, $M_{\mathcal{R}}(q, N)$ denotes the complexity of multiplication in $\mathcal{R}_q$. The row **Time(Param.)** denotes the resulting time complexity parametrized by the choice of some subroutines, and row **Time(Inst.)** reports the resulting complexity for a specific choice of said routines.

and the latter exceeds the intended threshold $2 \log(\lambda)^2 (\lambda - \log \lambda)$ for sufficiently large c . It remains to select q_2 , for which we can recycle the same instantiation strategy. This time, we assume q_1 fixed as in the previous point, and fix the input length to $\log(\lambda)^2$.

More precisely, for S , which has input length $l = \log(\lambda)^2$ and output modulus $p = q_1$, we recycle this strategy. We must find an LWE modulus q_2 and noise α_2 (e.g., $\alpha_2 = 2^{-\omega(\log(\lambda)^{1+d})}$ for a constant $d > c$) that satisfy the security condition from [26, Theorem 5.1]. The m^l term becomes $2^{O(\log(\lambda)^3)}$, which is dominated by the sub-exponentially small α_2 , ensuring suitable parameters exist.

Comparison with PRFs. As before, we compare our LWE-based instantiation with its full-fledged PRF counterpart. The main results are also summarized in Figure 6. Let us first focus on the runtime of our construction. We require $\log \lambda$ matrix multiplications for S' and $\log(\lambda)^2$ matrix multiplications for S . Let $\omega \in [2, 3]$ be an exponent depending on the choice of the matrix multiplication algorithm. The total cost is then $\mathcal{O}(\log(\lambda)^2 m^\omega M(q_2) + \log(\lambda) m^\omega M(q_1)) = \mathcal{O}(\log(\lambda)^2 m^\omega M(q_2))$. On top of this, we should add the cost of evaluating B , which amounts to $\mathcal{O}(dM(Q)) = \mathcal{O}(\log(\lambda)^2 M(Q))$. The total is $\mathcal{O}(\log(\lambda)^2 m^\omega M(q_2) + \log(\lambda)^2 M(Q))$. The key size is instead $\mathcal{O}(m \log q_2)$.

Let us now turn to the full PRF. For this comparison, we consider a BLMR PRF that processes inputs of length λ . At this point, we stumble upon a crucial issue. The reduction provided by [26], and summarized in Theorem 18, falls short when it comes to inputs whose length is linear in λ . More precisely, the resulting α would be too short to claim the hardness of the corresponding LWE instance. This confirms that Construction 3 requires less structured building blocks and, as in this case, extends the toolset available to crypto-scheme designers by enabling the use of blocks that would otherwise be unfit.

7 Implications on the Query Complexity of MACs

In this section, we briefly discuss the query complexity of MACs derived from our constructions, defining query complexity as the maximum number of calls the KGen^G , Mac^G , and Vrfy^G algorithms make to an underlying primitive G while maintaining uf-cma -security. We restrict ourselves to complexity with respect to pseudorandom generators, one-way permutations, and one-way functions.

The Goldreich–Goldwasser–Micali (GGM) construction [36] in combination with Levin’s domain extension technique [47] yields a pseudorandom function with $\text{poly}(\lambda)$ input length through $\omega(\log \lambda)$ calls to a length-doubling PRG. This in turn implies an analogous result on the query complexity of PRFs with respect to one-way functions due to the work of Haitner, Reingold, and Vadhan [41] and Vadhan and Zheng [70], who showed that PRGs allow for a black-box construction from one-way functions in $\tilde{\mathcal{O}}(\lambda^3)$ calls. Similarly, due to the Goldreich-Levin Theorem [39], which implies a construction of PRGs from one-way permutations with exactly one call, we can derive a result on the query complexity with respect to the latter. Since the Carter–Wegman–Shoup paradigm yields secure MACs when instantiated with a PRF, said query bounds immediately translate to analogous upper bounds on the query complexity of MACs with respect to PRGs, one-way functions, and one-way permutations.

Construction 3 instantiated as in Theorem 7 yields a uf-cma secure MAC. The problem of achieving small-domain 1-adaptive weak PRFs from PRGs remains open, and, for the time being, we take S and S' to be full PRFs. Using the GGM construction for S and S' , and a $\log(\lambda)^2$ -wise independent function for C (which requires no G calls), the total number of queries to G is $\mathcal{O}(\log(\lambda)^2)$. This matches exactly the query complexity resulting from the GGM-Levin strategy.

Recent work [9] proves that, if we restrict ourselves to *tree* constructions, a framework generalizing the GGM trees, no black-box construction making $\mathcal{O}(\log \lambda)$ queries to PRGs (but having input length $\omega(\log \lambda)$) can exist. The result is stronger, and holds even for *weak* PRFs. Therefore, it also applies to 1-adaptive weak PRFs. In this sense, our construction achieves *quasi*-optimal PRG query complexity — further improvements require a paradigm that fundamentally diverges from tree constructions (if it exists).

It is also interesting to examine the query complexity in relation to weaker primitives. For example, Maurer and Tessaro’s cascade construction [52] implies that PRFs with a $\log(\lambda)^2$ input length can be constructed from λ -bounded weak PRFs (a type of weak PRF that is pseudorandom for a limited number of calls) in $\mathcal{O}(\log \lambda)$ many calls. Due to our construction, the same bound applies immediately to 1-adaptive weak PRFs.

We also observe that weak and full PRFs are subject to the same lower bound on query complexity with respect to PRGs. However, in concrete instantiations based on computational assumptions, such as DDH and LWE, weak PRFs are often significantly simpler and faster. As an intermediate class bridging between the two, we conjecture that a similar speedup is possible for 1-adaptive weak PRFs through instances based on ad-hoc computational assumptions. Construction 3 and Theorem 7 suggest a sufficient strategy to achieve said speedup, i.e., finding lightweight instances of small-domain 1-adaptive weak PRFs and small-domain 1-adaptive bounded weak PRFs, and plugging them into Construction 3. Identifying lightweight instances of small-domain 1-adaptive weak PRFs under standard cryptographic assumptions constitutes a compelling direction for future research.

A Preliminaries

Notation. For a finite set S , we denote sampling an element x uniformly from S by $x \leftarrow \$ S$. For two (bit-)strings x and y , we denote concatenation by $x\|y$. If x and y are of the same length, then $x \oplus y$ denotes their bitwise XOR. We call a function $f: \mathbb{N} \rightarrow \mathbb{R}^+$ negligible in n if for every $k \in \mathbb{N}$ there exists $n_0 \in \mathbb{N}$ such that $|f(n)| \leq n^{-k}$ holds for every $n \geq n_0$. We use negl to denote an arbitrary negligible function. The security parameter is λ , and when referring to a function as negligible, we mean negligible in λ . We say that an event occurs with overwhelming probability if it occurs with probability at least $1 - \text{negl}(\lambda)$ for any negligible function negl . Let $\Sigma = \{0, 1\}$ be the binary alphabet. We denote Σ^n by $\mathcal{D}_n$ for any $n \in \mathbb{N}$, where $|\mathcal{D}_n| = 2^n$ denotes its size. Let $\mathcal{D}^* := \bigcup_{n \geq 0} \Sigma^n = \Sigma^*$ be the space of all finite binary strings. Correspondingly, let $\mathcal{D}_\lambda$ denote the security space with key $k \in \mathcal{D}_\lambda$. Let $\mathcal{D}_X$, $\mathcal{D}_R$, and $\mathcal{D}_Y$ denote the preimage, randomness, and image spaces, respectively, with standard variable names $x \in \mathcal{D}_X$, $r \in \mathcal{D}_R$, and $y \in \mathcal{D}_Y$, which we use consistently throughout the paper unless stated otherwise. We write $\text{Func}_{\mathcal{D}_Y}^{\mathcal{D}_X}$ to denote the set of functions mapping from $\mathcal{D}_X$ to $\mathcal{D}_Y$, where the superscript indicates the codomain and the subscript the domain. We simplify this notation to Func_y^x using the common variable names. We assume all functions and samplings to be efficient, i.e., computable in polynomial time, unless stated otherwise. We use the code-based game-playing framework of Bellare and Rogaway [17]. Here, an (efficient) adversary $\mathcal{A}$ always considers an arbitrary *probabilistic polynomial time* PPT algorithm. We denote $\mathcal{A}^{\mathcal{O}_1, \dots, \mathcal{O}_n}(x_1, \dots, x_m) = b$ as the execution of $\mathcal{A}$ with *Oracle* access to $\mathcal{O}_1, \dots, \mathcal{O}_n$ using inputs $x_1, \dots, x_m$, and output b . The notation $x \leftarrow \$ \mathcal{A}$ emphasizes sampling from the set of all possible outputs of $\mathcal{A}$.

Pseudorandom Functions. The pseudorandomness of PRFs is classically defined via an indistinguishability game in which a distinguisher D attempts to distinguish between a keyed pseudorandom function F and a truly random function f . Formally, a distinguisher D is given oracle access either to $F(k, \cdot)$, where $k \in \mathcal{D}_\lambda$ is sampled uniformly at random and determines the function $F(k, \cdot)$ from the set $\text{Func}_y^{\lambda, x} = \{F \mid F: \mathcal{D}_\lambda \times \mathcal{D}_X \rightarrow \mathcal{D}_Y\}$, or to a function f drawn uniformly from the set $\text{Func}_y^x = \{f \mid f: \mathcal{D}_X \rightarrow \mathcal{D}_Y\}$. We say that a function $F: \mathcal{D}_\lambda \times \mathcal{D}_X \rightarrow \mathcal{D}_Y$ is *pseudorandom* if for all efficient distinguishers D there exists a negligible function negl such that

$$\left| \Pr \left[D^{F(k, \cdot)}(1^\lambda) = 1 \right] - \Pr \left[D^{f(\cdot)}(1^\lambda) = 1 \right] \right| \leq \text{negl}(\lambda),$$

where the first probability is over a uniform sampling of $k \in \mathcal{D}_\lambda$ and the randomness of D , while the second probability is over a uniform sampling of $f \in \text{Func}_y^x = \{f \mid f: \mathcal{D}_X \rightarrow \mathcal{D}_Y\}$ and the randomness of D .

Non-adaptive pseudorandom functions are a relaxation of the standard PRF security experiment, where the distinguisher is restricted in how it may query its oracle. More precisely, we consider a *non-adaptive distinguisher* D , which is an efficient algorithm that must select all of its polynomial many q queries $x_1, \dots, x_q$ *before* receiving any of the corresponding output values $y_1, \dots, y_q$, and then outputs its decision bit.

Weak Pseudorandom Functions. We observe that in the PRFs security experiment, the distinguisher can choose the input queries adaptively. In contrast, for *weak pseudorandom functions* wPRF, a distinguisher D can not control any input query. The distinguisher only obtains a transcript for polynomial many q input-output pairs, where the input is chosen independently and

uniformly at random. Formally, we call $\mathbf{wF}: \mathcal{D}_\lambda \times \mathcal{D}_X \rightarrow \mathcal{D}_Y$ a weak pseudorandom function if for all efficient distinguishers D , there exists a negligible function negl such that

$$\left| \Pr[D(1^\lambda, \{x_i, \mathbf{wF}(k, x_i)\}_{i=1}^q) = 1] - \Pr[D(1^\lambda, \{x_i, f(k, x_i)\}_{i=1}^q) = 1] \right| \leq \text{negl}(\lambda),$$

where $\{x_i\}_{i=1}^q$ is a set of size q including uniformly sampled $x_i \in \mathcal{D}_X$, the first probability is over a uniform sampling of $k \in \mathcal{D}_\lambda$, and the randomness of D , while the second probability is over a uniform sampling of $f \in \text{Func}_Y^x$, and the randomness of D . Further, we highlight the fact that for the wPRF experiment, it makes no difference whether the D gets all q outputs at once or one by one.

Strong Pseudorandom Permutations. Let $\text{Perm}_x^x = \{\pi \mid \pi: \mathcal{D}_X \rightarrow \mathcal{D}_X\}$ denote the set of 1-to-1 functions over $\mathcal{D}_X$. For any $\pi \in \text{Perm}_x^x$, we let π^{-1} denote its inverse. Let $\mathbf{F}: \mathcal{D}_\lambda \times \mathcal{D}_X \rightarrow \mathcal{D}_X$ denote a keyed function such that for each $k \in \mathcal{D}_\lambda$, $\mathbf{F}(k, \cdot)$ is a 1-to-1 mapping over $\mathcal{D}_X$. Similar to π , let $\mathbf{F}^{-1}(k, \cdot)$ denote its inverse. We say that $\mathbf{F}$ is a strong pseudorandom permutation if for any PPT distinguisher D there exists a negligible function $\text{negl}(\lambda)$ such that

$$\left| \Pr[D^{\mathbf{F}(k, \cdot), \mathbf{F}^{-1}(k, \cdot)}(1^\lambda) = 1] - \Pr[D^{\pi(\cdot), \pi^{-1}(\cdot)}(1^\lambda) = 1] \right| \leq \text{negl}(\lambda).$$

The first probability is over a uniform sampling of $k \in \mathcal{D}_\lambda$ and the randomness of D , while the second probability is over a uniform sampling of $\pi \in \text{Perm}_x^x$ and the randomness of D .

Hash Functions.

A (keyed) *hash function* $H: \mathcal{D}_\lambda \times \mathcal{D}^* \rightarrow \mathcal{D}_Y$ compresses input of arbitrary length to output with a fixed length. Formally, H represents a family of functions, where each key $k \in \mathcal{D}_\lambda$ instantiates a specific function $h_k: \mathcal{D}^* \rightarrow \mathcal{D}_Y$ defined as $h_k(x) := H(k, x)$. The properties defined below are probabilistic statements over the uniform random choice of the key $k \leftarrow \$ \mathcal{D}_\lambda$, which is equivalent to sampling a function h_k uniformly from the family.

A common property of hash functions is *collision resistance*, which states that finding two preimages leading to the same image for a hash function with a fixed key is negligible for any efficient adversary. *Universality* generalizes the definition of collision resistance.

Definition 8 (Universality [71]). *We call $H: \mathcal{D}_\lambda \times \mathcal{D}^* \rightarrow \mathcal{D}_Y$ universal if for any $x_1, x_2 \in \mathcal{D}^*$ with $x_1 \neq x_2$, we have*

$$\Pr[H(k, x_1) = H(k, x_2)] = \frac{1}{|\mathcal{D}_Y|}$$

with a uniform random key $k \leftarrow \$ \mathcal{D}_\lambda$ instantiating the hash function H .

In contrast, *strong universality* (or pairwise independence [32]) states that for a hash function (with a fixed key), the images of any two distinct preimages are independently (uniformly) randomly distributed.

Definition 9 (Strong Universality [71]). *We call $H: \mathcal{D}_\lambda \times \mathcal{D}^* \rightarrow \mathcal{D}_Y$ strongly universal if for any $x_1, x_2 \in \mathcal{D}^*$ with $x_1 \neq x_2$ and any $y_1, y_2 \in \mathcal{D}_Y$, we have*

$$\Pr[H(k, x_1) = y_1 \wedge H(k, x_2) = y_2] = \frac{1}{|\mathcal{D}_Y|^2}$$

with a uniform random key $k \leftarrow \$ \mathcal{D}_\lambda$ instantiating the hash function H .

In contrast, *XOR-universality* is a specification of strong universality, which states that for a hash function (with a fixed key), the XOR of the images of any two distinct preimages is (uniformly) randomly distributed.

Definition 10 (XOR-Universality [68]). *We call $H: \mathcal{D}_\lambda \times \mathcal{D}^* \rightarrow \mathcal{D}_Y$ XOR-universal if for any $x_1, x_2 \in \mathcal{D}^*$ with $x_1 \neq x_2$ and any $\delta \in \mathcal{D}_Y$, we have*

$$\Pr[H(k, x_1) \oplus H(k, x_2) = \delta] = \frac{1}{|\mathcal{D}_Y|}$$

with a uniform random key $k \leftarrow \$ \mathcal{D}_\lambda$ instantiating the hash function H .

If $\mathcal{D}_Y = \{0, 1\}^\ell$ for some $\ell \in \mathbb{N}$, then $\oplus$ represents the standard XOR of bitstrings. Otherwise, it should be understood as the operation of an additive group $(\mathcal{D}_Y, \oplus)$. The above definitions form a hierarchy, where strong universality is the strongest property and (weak) universality is the weakest [71, 68]. That is, being strongly universal implies being XOR-universal, which in turn implies weak universality. Finally, all three definitions come with slight relaxations. We say that H is ϵ -almost universal (with $\epsilon > 0$) if the aforementioned collision probability satisfies $\Pr[H(k, x_1) = H(k, x_2)] = \epsilon/|\mathcal{D}_Y|$. The idea is straightforwardly generalized to XOR-universal and strongly universal functions.

Strong universality is sometimes referred to as *pairwise independence*. We can generalize the concept to t -wise independence for any $t \in \mathbb{N}$ as follows.

Definition 11 (t -wise Independence). *A function $H: \mathcal{D}_\lambda \times \mathcal{D}^* \rightarrow \mathcal{D}_Y$ is t -wise independent if for any pairwise distinct $x_1, \dots, x_t \in \mathcal{D}^*$ and any $y_1, \dots, y_t \in \mathcal{D}_Y$, we have*

$$\Pr[H(k, x_1) = y_1 \wedge H(k, x_2) = y_2 \wedge \dots \wedge H(k, x_t) = y_t] = \frac{1}{|\mathcal{D}_Y|^t}$$

with a uniform random key $k \leftarrow \$ \mathcal{D}_\lambda$ instantiating the hash function H .

B Preliminaries on Lattice-Based Cryptography

Lattices. A lattice Λ is any discrete additive subgroup of $\mathbb{R}^m$. Any lattice can be represented as $B \cdot \mathbb{Z}^n$, for some matrix $B \in \mathbb{R}^{m \times n}$ whose columns are linearly independent and $n \leq m$. B is a *basis* of Λ , and is not necessarily unique.

We define for integers $\ell \geq 1$ the $\|\cdot\|_\ell$ as

$$\|v\|_\ell = \left(\sum_i |v_i|^\ell \right)^{1/\ell}$$

and $\|v\|_\infty = \max_i |v_i|$. For each lattice and $\ell \in \mathbb{N} \cup \{+\infty\}$ we define the minimum distance in the ℓ -norm as

$$\lambda_1^\ell(\Lambda) = \min_{v \in \Lambda \setminus \{0\}} \|v\|_\ell,$$

i.e., the norm of the shortest non-zero vector in Λ . The GapSVP_γ parametrized by $\gamma \geq 1$ is a decisional promise problem which asks, given an input basis B and rational $r > 0$, to determine whether the lattice Λ generated by B is such that $\lambda_1^2(\Lambda) \leq r$ or $\lambda_1^2(\Lambda) \geq \gamma r$. GapSVP_γ is conjectured

to be worst-case hard for polynomial and subexponential γ . Thanks to the celebrated worst-to-average case reductions of Regev [66] and Peikert [62], later improved by [29, 53] it is known that the hardness of GapSVP_γ implies the average-case hardness of the celebrated Learning With Errors problem, defined as follows.

Definition 12 (Decisional Learning With Errors). *Fix integer parameters m, n, q , and a distribution χ supported over $\mathbb{Z}_q^m$. Let $A \leftarrow \$ \mathbb{Z}_q^{m \times n}$, $s \leftarrow \$ \mathbb{Z}_q^n$, $e \leftarrow \chi$, $u \leftarrow \$ \mathbb{Z}_q^m$. The decisional Learning With Errors problem with parameters (m, n, q, χ) asks to distinguish $(A, As + e)$ from (A, u) .*

The aforementioned reductions are crucially affected by the choice of (m, n, q, χ) , i.e., for some parameters LWE is known to be tractable, while for others it is either covered by reductions, or not known to be efficiently solvable.

Learning With Rounding. Despite the randomized nature of the LWE problem, there exist established techniques to derandomize it while still ensuring pseudorandomness and search hardness. Define a function $\lfloor \cdot \rfloor_p : \mathbb{Z}_q \rightarrow \mathbb{Z}_p$ as

$$\lfloor x \rfloor_p = \left\lfloor \frac{px}{q} \right\rfloor.$$

We extend $\lfloor \cdot \rfloor_p$ to vectors by applying it component-wise. The main idea is that by applying $\lfloor \cdot \rfloor_p$ to LWE samples, the modest magnitude of the LWE error allows us to neglect the latter, thus leading to deterministically generated samples that remain pseudorandom. Formally, we use the following version (decisional) LWR problem.

Definition 13 (Learning With Rounding). *Let $A \leftarrow \$ \mathbb{Z}_q^{m \times n}$, $s \leftarrow \$ \mathbb{Z}_q^n$, $u \leftarrow \$ \mathbb{Z}_p^m$. The decisional LWR problem asks to distinguish $(A, \lfloor As \rfloor_p)$ from (A, u) .*

LWR inherits its hardness from LWE for appropriate parameters [8, 6, 4, 25, 1, 45], which enables the construction of deterministic lattice-based pseudorandom functions. Nevertheless, we hereby focus on applications of LWR for weak pseudorandom functions. Assume LWR is hard for some parameter choice, and consider the function $F : \mathbb{Z}_q^n \times \mathbb{Z}_q^n \rightarrow \mathbb{Z}_p$ defined as

$$F_k(x) = \lfloor \langle x, k \rangle \rfloor_p$$

The latter is a weak PRF under the assumed worst-case hardness of some lattice problems.

C Relevant Hashes and PRFs

C.1 Universal Hashes

Strongly Universal Hash over $\mathbb{Z}_p$. For a prime p , define a keyed function H with key space $\mathbb{Z}_p^2$, and input and output space both equal to $\mathbb{Z}_p$. Given a uniformly random key (a, b) , we set for $x \in \mathbb{Z}_p$

$$H_{a,b}(x) = ax + b.$$

The function is strongly universal by the following reasoning: consider two distinct points $x_1 \neq x_2$; then, for any two $y_1, y_2 \in \mathbb{Z}_p$, there exists a unique polynomial $f \in \mathbb{Z}_p[X]$ such that $\deg(f) = 1$ and $f(x_i) = y_i$ for $i \in \{1, 2\}$; the probability that a and b equal the coefficients of f is exactly $1/p^2$.

Strongly Universal Hash over $\mathbb{G}$. Let $\mathbb{G}$ be a cyclic group of prime order p where the DDH assumption holds. Define a keyed function H with keyspace $\mathbb{Z}_p \times \mathbb{G}$, and input and output space equal to $\mathbb{G}$. Given a uniformly random key (a, b) we set for each $x \in \mathbb{G}$

$$H_{a,b}(x) = ax + b.$$

To show that H is strongly universal, consider two $x_1 \neq x_2$. Then, for any two $y_1, y_2 \in \mathbb{G}$ we consider the system

$$\begin{cases} ax_1 + b = y_1 \\ ax_2 + b = y_2 \end{cases}$$

in the unknowns a and b . To solve for a , we subtract both equations getting $a(x_1 - x_2) = y_1 - y_2$. Since $x_1 - x_2 \neq 0$, and $\mathbb{G}$ is prime and has prime order, by Lagrange's theorem, $x_1 - x_2$ must be a generator of $\mathbb{G}$. It follows that there exists a unique a fulfilling the previous requirement. Then, b is the unique value $y_1 - ax_1 = y_2 - ax_2$, which is always guaranteed due to the previous condition on a . Thus, the pair (a, b) is unique, and has probability $1/p^2$ of being sampled.

XOR-Universal Hash Over $\{0, 1\}^m$. For any fixed $m \in \mathbb{N}$, define a keyed function H with keyspace $\{0, 1\}^m$, and input and output space both equal to $\{0, 1\}^m$. Given a uniformly random key $a \leftarrow \{0, 1\}^m$, we set for $x \in \{0, 1\}^m$

$$H_a(x) = a \cdot x,$$

where the $\cdot$ operator should be understood as the product over $\mathbb{F}_{2^m}$ of the elements corresponding to bitstrings a and x . Note that for distinct $x_1 \neq x_2$ and arbitrary $\delta \in \{0, 1\}^m$ it holds

$$\Pr[(ax_1 \oplus ax_2) = \delta] = \Pr[a(x_1 \oplus x_2) = \delta] = \Pr[a = (x_1 \oplus x_2)^{-1}\delta] = 1/2^m.$$

Note that $x_1 - x_2$ is always nonzero, and therefore guaranteed to be invertible.

C.2 Weak PRFs

Learning With Rounding Weak PRF. Let $n \in \mathbb{N}$ be integers, and $q > p$ be two moduli. For $x \in \mathbb{Z}_q$, define the function $\lfloor \cdot \rfloor_p : \mathbb{Z}_q \rightarrow \mathbb{Z}_p$ as

$$\lfloor x \rfloor_p = \lfloor x \cdot p/q \rfloor \quad \forall x \in \mathbb{Z}_q.$$

Given a key $k \in \mathbb{Z}_q^n$, and an input $x \in \mathbb{Z}_q^n$, we define

$$\text{wF}(k, x) = \lfloor \langle k, x \rangle \rfloor_p.$$

DDH Weak PRF. Let $\mathbb{G}$ be a cyclic group of prime order p where the DDH assumption holds. Sample a uniformly random key $k \leftarrow \mathbb{Z}_p$. On input $x \in \mathbb{G}$, compute⁵

$$\text{wF}(k, x) = x^k.$$

⁵We use the additive notation, as we find it more convenient to state the impossibility results of section 5, for which the DDH weak PRF is needed.

C.3 PRFs

Naor-Reingold PRF. Let $\mathbb{G}$ be a cyclic group of prime order p and generator g where the DDH assumption holds. The popular Naor-Reingold PRF requires a key $\vec{k} \leftarrow \mathbb{Z}_p^{m+1}$ to map an input $x \in \{0, 1\}^m$ into a pseudorandom element of $\mathbb{G}$. Concretely, it computes

$$F(\vec{k}, x) = g^{k_0 \prod_{i=1}^m k_i^{x_i}}.$$

BLMR PRF [26]. Let n, m, q, p be integers, with $p \mid q$ and $m = n \lceil \log q \rceil$. Let $\eta_{Bin}(m)$ be a distribution parametrized by m , with support over $\mathbb{Z}^m$. The BLMR PRF uses two public matrices $A_0, A_1 \in \mathbb{Z}_q^{m \times m}$ having each row sampled independently from $\eta_{Bin}(m)$ such that both matrices are full rank. The secret key $\vec{k}$ is a uniformly random element of $\mathbb{Z}_q^m$. On input $x \in \{0, 1\}^\ell$, it computes

$$F(\vec{k}, x) = \left[\prod_{i \in [\ell]} A_{x_i} \cdot \vec{k} \right]_p \in \mathbb{Z}_p^m.$$

Its hardness stems from that of LWE according to the following theorem.

Theorem 18 ([26]). *The above function is pseudorandom if LWE with parameters (q, n, χ_α) is hard, with χ_α denoting the following distribution parametrized by real $\alpha \in (0, 1)$:*

1. Sample $X \leftarrow \mathcal{N}(0, \alpha/\sqrt{2\pi})$.
2. Return $\lceil qX \rceil \bmod q$.

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